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Bridging game theory and multi-agent systems: Development status and future prospects

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ABSTRACT

The integration of game theory and multi-agent systems (MASs) has been systematically examined as a transformative paradigm for modeling strategic interactions among autonomous entities in advanced technological systems. This paper focuses on the synergy between game-theoretic principles and MASs, with emphasis on their applications to complex operational domains such as autonomous system coordination, distributed system control, and intelligent network management. Firstly, foundational concepts, historical developments, and classifications of both fields have been analyzed. The analysis highlights how game theory provides robust frameworks for addressing challenges in cooperative control, resource allocation, and swarm dynamics within advanced operational contexts. For instance, game-theoretic approaches to swarm-vs.-swarm engagement in contested environments and distributed guidance for interception systems have been investigated. Subsequently, key application scenarios have been explored, including robust path optimization for autonomous agents operating under uncertain conditions, such as GPS-denied or similar challenging environments. Challenges unique to complex applications, such as high-dimensional state spaces, real-time computational demands, and communication constraints in dynamic environments, have also been identified. Finally, future research directions emphasize the development of scalable distributed algorithms, enhancement of resilience against adversarial disruptions, and optimization of decision-making under incomplete information—critical for advancing autonomous systems in diverse technological fields. Overall, this paper offers a comprehensive analysis of the application of game theory in MASs and anticipates future advancements in the field.

Contents

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1. Introduction

Game theory, rooted in mathematics and economics, provides a rigorous framework for modeling strategic interactions among rational decision-makers [1]. Its principles are particularly relevant in the context of multi-agent systems (MASs), which emulate the collective intelligence observed in natural systems such as ant colonies and bird flocks [2,3]. MASs focus on designing autonomous agents that collaborate or compete to achieve shared objectives, offering a powerful paradigm for solving complex problems beyond the capabilities of individual entities. The synergy between game theory and MASs not only deepens our understanding of agent dynamics but also drives the development of advanced decision-making models and algorithms tailored for intricate environments [4].

In an increasingly interconnected world, the need to model and analyze strategic interactions among autonomous entities has become paramount. From autonomous vehicles navigating urban landscapes to drones coordinating search-and-rescue missions, and from resource allocation in smart grids to swarm robotics in industrial automation, achieving seamless coordination among decision-making entities requires a profound understanding of strategic interactions [5–8]. The integration of game theory and MASs provides a foundational framework to address these challenges, enabling systems to make optimal decisions individually while coordinating effectively in dynamic and complex environments [9–12].

Despite its transformative potential, the practical application of game theory in MASs faces significant challenges. Computational complexity escalates exponentially with the number of agents, particularly in large-scale systems [13], necessitating efficient algorithms to solve game-theoretic models. Incomplete information introduces uncertainty into decision-making, raising critical questions about how to derive optimal strategies under such conditions [14]. Additionally, ensuring system stability in dynamically changing environments remains a pressing concern, requiring robust adaptive solutions [15,16]. However, these challenges also present opportunities for innovation. Advances in distributed computing [17], reinforcement learning [18], and heuristic optimization offer promising avenues to address computational complexity and incomplete information [19]. Furthermore, the development of novel protocols and methodologies can enhance system stability and scalability [20–22].

This paper systematically reviews the integration of game theory and MASs, highlighting key application scenarios such as cooperative control, cooperative guidance, resource allocation, path optimization, swarm-vs.-swarm engagement, and search and capture. Through a comprehensive analysis of these scenarios, we demonstrate how game theory provides effective strategies and solutions, thereby enhancing the performance and efficiency of MASs. The main contributions of this study are summarized as follows.

(1) A systematic review of the fundamental theories and applications of game theory and MASs are presented. By analyzing their core concepts, historical development, and classifications, this paper establishes a comprehensive understanding of their integration. Furthermore, the mapping relationships between the elements of game theory and MASs are explored, highlighting the methods and advantages of using game theory to solve multi-agent problems.
(2) Key application scenarios where game theory and MASs demonstrate significant potential are investigated, including cooperative control, cooperative guidance, resource allocation, path optimization, swarm-vs.-swarm engagement, and search and capture. Through case studies and research findings, the paper illustrates how game theory enhances the performance of MASs in these domains.
(3) The primary challenges in the practical application of game theory in MASs are identified, such as computational complexity, incomplete information, and system stability. In response, future research directions are proposed, including the development of distributed algorithms, optimization under incomplete information, and enhancement of system robustness. These insights provide valuable guidance for advancing the field.

The paper is organized as follows. Section 2 introduces game theory, covering its fundamentals, history, and classifications. Section 3 discusses MASs, including their development, components, and architectures. The integration of game theory in MASs are then examined in Section 4, highlighting key correlations. Application scenarios of game theory in MASs are explored in Section 5. Section 6 identifies challenges and future directions. Section 7 concludes with key findings and emphasizes the importance of interdisciplinary research advancement.

Review methodology. This review adopts a hybrid methodology combining systematic literature search with expert-driven thematic analysis to ensure comprehensive coverage while maintaining conceptual coherence. The literature search was conducted across three major databases including Scopus, Web of Science, and IEEE Xplore, covering publications from 2000 to 2025. Search queries employed Boolean combinations of domain-specific keywords related to game-theoretic concepts, multi-agent system architectures, and aerospace applications. After removing duplicates and applying inclusion criteria including peer-reviewed journals or conferences and explicit integration of game-theoretic models with MAS implementation, more than 200 seminal works were selected for in-depth analysis. Selection prioritized methodological innovation, aerospace and robotics applications with experimental validation, and foundational theoretical contributions. To complement database searches and capture influential early works that may predate digital indexing or use alternative terminology, backward snowballing was performed on highly cited papers and seminal references identified through expert knowledge of the field.

2. Overview of game theory

2.1. Concept

Game theory employs mathematical models to analyze strategic interactions among rational decision-makers, where each player has individual objectives and their choices are interdependent with the actions of others [23]. As a branch of complex system analysis, it has found extensive application in various fields, including economics, social sciences, computer science, and operational research, providing insights into decision-making processes where outcomes hinge on the choices of multiple parties [24].

2.2. Development

In game theory, scholars have collectively developed a theoretical framework enabling the analysis of complex interactions among agents, as shown in Table 1.

The n-person game framework developed by John von Neumann and Oskar Morgenstern laid the foundation for analyzing strategic interactions among multiple participants in game theory [25]. John Nash introduced Nash equilibrium concept [26], which describes a state in non-cooperative games where each player has chosen the best strategy and has no incentive to unilaterally deviate. This concept enables the prediction of agent behavior in both competitive and cooperative settings and the achievement of a stable outcome. The Shapley value [27], proposed by Lloyd Shapley, provides a method for distributing total payoffs in cooperative games, ensuring that each participant's share corresponds to their marginal contribution. This approach facilitates the fair distribution of collaborative task outcomes, ensuring that each agent's reward reflects their contribution.

Rufus Isaacs made significant contributions to differential games [28], which model continuous changes in players' decisions and outcomes over time, enabling the simulation and analysis of agent interactions in continuous-time settings. Robert Aumann extended the concept of Nash equilibrium by considering a broader set of strategies [29], demonstrating its stability even when players may adopt irrational strategies, thereby providing a more robust analysis of stability in uncertain environments. John Harsanyi developed the framework of games with incomplete information and introduced the Bayesian Nash equilibrium [30], where players have uncertain knowledge about others' types or preferences, allowing for the analysis of decision-making under uncertainty. Maynard Smith and George Price co-founded evolutionary game theory [31], introducing the concept of evolutionarily stable strategies to analyze the evolution and stability of strategies in biological populations. These concepts are also applied to model the strategic behavior and stable states in competitive and cooperative scenarios among agents. Furthermore, Robert Aumann explored the concept of common knowledge [32], where all participants are aware of a fact and mutually aware of each other's knowledge, facilitating the analysis of coordination and trust in decision-making processes.

David Kreps et al. [33] developed the reputation model and sequential equilibrium to analyze decision-making based on reputation in repeated interactions. These models explore how agents establish and maintain reputations over time and how reputations influence their strategic choices. Drew Fudenberg and Jean Tirole introduced the perfect Bayesian equilibrium [34], a concept for dynamic games where participants update their beliefs using Bayes' rule and choose optimal strategies at each stage. In MASs, perfect Bayesian equilibrium is useful for analyzing decision-making processes in dynamic environments, especially when uncertainty and information updates are involved.

Matthew Jackson systematically introduced network game models [35], which helped to understand how individuals strategically interact within these networks to form complex phenomena and how these network structures affect the stability and efficiency of MASs. Algorithmic game theory [36], an intersection of game theory and computer science, was explored by researchers including Noam Nisan, who engaged in the design and analysis of algorithms for game-theoretic problems and investigated how computational constraints impact agent behavior in game-theoretic environments. This field offered a theoretical foundation and analytical tools for MASs, assisting researchers in comprehending and designing agent behavior in complex and dynamic environments.

Table 1
Evolution and analytical characteristics of game theory.

Period	Contributor(s)	Contributions	Characteristics
1940–1959	Von Neumann and Morgenstern	N-person game structures [25]: https://doi.org/10.1515/9781400829460	Theoretical foundations of multi-player games
	John Nash	Nash equilibrium and Nash theorem [26]: https://doi.org/10.1073/pnas.36.1.48	Stable strategies in non-cooperative games
	Lloyd Shapley	Shapley value [27]: https://doi.org/10.1515/9781400881970-018	Evaluation and allocation of contributions in cooperative games
1960–1979	Rufus Isaacs	Differential game [28]: https://doi.org/10.2307/3613661	Continuous-time dynamic systems
	Robert Aumann	Strong Nash equilibrium [29]: https://doi.org/10.2307/2171725	Nash equilibrium under coalition consistency
	John Harsanyi	Incomplete information games and Bayesian Nash equilibrium [30]: https://doi.org/10.1287/mnsc.14.3.159	Uncertainty about other players' types or preferences
	Maynard Smith and George Price	Evolutionary game theory and evolutionarily stable strategy [31] https://doi.org/10.1038/246015a0	Dynamic process of swarm evolution over time
	Robert Aumann	Common knowledge [32]: https://doi.org/10.1007/978-3-319-20451-2_40	Game context and information assumptions
1980–1999	David Kreps, Paul Milgrom, and Robert Wilson	Reputation model and sequential equilibrium [33]: https://doi.org/10.1016/0022-0531(82)90029-1	Decision-making in repeated interactions
	Drew Fudenberg and Jean Tirole	Perfect Bayesian equilibrium [34]: https://doi.org/10.1016/0022-0531(91)90155-W	Dynamic decision-making with incomplete information and revision
2000–Now	Matthew Jackson	Social and economic networks [35]: https://doi.org/10.2307/j.ctvcm4gh1	Game behavior in network structures
	Noam Nisan, Tim Roughgarden, Eva Tardos, and Vijay Vazirani	Algorithmic game theory [36]: https://doi.org/10.1017/CBO9780511800481	Design and analysis of algorithms for game-theoretic problems

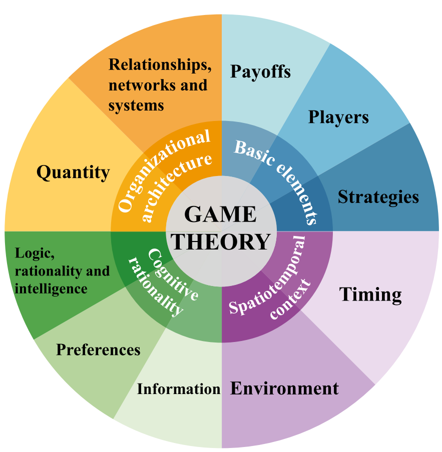

Fig. 1. Elements of game theory.

2.3. Elements

Game theory provides a structured framework for modeling strategic interactions among rational actors in diverse contexts [37], offering insights into decision-making processes and the resultant outcomes from these interactions.

Game theory revolves around several key elements that define the structure of the games being analyzed. The following discussion will explore these fundamental constructs, the context in which the games take place, the rational decision-making that guides player behavior, and the structural framework that shapes their interactions, as illustrated in Fig. 1.

2.3.1. Basic elements

(1) Players. Represent the decision-making entities, often assumed to be rational, and can include a variety of actors such as individuals, enterprises, countries, network nodes, and system clusters. Each player aims to maximize their own payoff, pursuing goals that align with this objective.
(2) Strategies. Denotes the action plans that players can adopt during a game. The strategy set consists of all possible strategies available to a player.
(3) Payoffs. Numerical representations of the outcomes players receive after the game ends, such as profits, utilities, or satisfaction levels. They depend on the strategies chosen by all players, thereby influencing their strategic decisions [25,26,38].

2.3.2. Spatiotemporal context

(1) Timing. Game theory incorporates timing elements in strategic interactions, distinguishing between static, dynamic, repeated, and hybrid games. Static games involve simultaneous moves where the strategic environment remains constant, enabling a single, timeless analysis. Dynamic games, like chess, unfold over time, requiring examination of how strategies and outcomes evolve with each move. Repeated games involve successive interactions, with players' strategies adapting based on previous outcomes. Hybrid games combine elements of both static and dynamic interactions, creating a complex interplay of stable and changing components, akin to the varied pacing of real-world strategic scenarios [39].
(2) Environment. Environment can be categorized based on the level of certainty or uncertainty they present. In settings of complete certainty, outcomes are entirely predictable and can be modeled using deterministic approaches. Stochastic uncertainty introduces probabilistic elements, where outcomes are uncertain

but can be described through stochastic models. Interval uncertainty involves known ranges of possible values, addressed by interval models. Fuzzy uncertainty deals with vagueness and imprecision, handled effectively with fuzzy logic models. Rough uncertainty concerns the granularity of information, which is captured by rough models. Lastly, belief uncertainty involves subjective probabilities, modeled using belief models. Each of these uncertainty types necessitates distinct analytical methods to effectively navigate the complexities of the environment [40].

2.3.3. *Cognitive rationality*

(1) Information. In game theory, the distinction between complete and incomplete information is fundamental. In complete information games, players know all rules, strategies, and payoffs. In contrast, incomplete information arises when at least one player lacks full knowledge about others' payoffs, strategies, or rules. Additionally, games can have perfect information, where players know all past actions, as in chess, or imperfect information, involving uncertainty, such as in poker. These information variations significantly affect players' strategies and game complexity [41].

(2) Preferences. Player preferences are represented through various mechanisms: quantitative preferences assign numerical values to outcomes, vector preferences consider multiple criteria simultaneously, complete order preferences rank all outcomes in a clear hierarchy, and partial order preferences allow for some outcomes to be equally preferred, offering a flexible ranking system. Understanding these preference types is essential for analyzing decision-making processes in game theory [42].

(3) Logic, rationality and intelligence. Game theory examines strategic interactions among rational decision-makers, assuming perfect intelligence and infinite reasoning, where participants process information flawlessly and predict each other's actions without limits. Conversely, limited intelligence and finite reasoning reflect more realistic scenarios, acknowledging bounded cognitive abilities and potential errors in predicting others' strategies. Complete information and shared knowledge ensure participants are fully informed and have a common understanding, fostering cooperation and trust. Conversely, incomplete information and a lack of shared knowledge increase uncertainty, as participants may misjudge others' knowledge or intentions, complicating strategic analysis [43].

2.3.4. *Organizational architecture*

(1) Relationships, networks and systems. Game theory examines various types of relationships among players, including non-cooperative and cooperative dynamics. Non-cooperative games involve players acting independently, each pursuing their own interests without formal agreements. Cooperative games, in contrast, involve players forming binding agreements to achieve common goals. Network-type games focus on interconnected players and the flow of information and strategies through these networks. Systemic-type games analyze how individual interactions shape overall system behavior, emphasizing interdependencies in complex strategic systems [44].

(2) Quantity. In game theory, quantitative aspects such as the number of players, the size of strategy sets, the structure of relationships among players, and the depth of reasoning are fundamental. These quantities can be finite or infinite, and their nature significantly influences the complexity of the game and the strategic analysis required. For example, games with a finite number of players and a limited strategy set are often easier to model and predict, whereas infinite settings introduce greater complexity. Similarly, whether players engage in finite or infinite levels of reasoning affects their interactions and the potential outcomes of the game. Understanding these quantitative dimensions is crucial for applying game theory to real-world scenarios, ranging from small-scale interactions to large, dynamic systems [45].

2.4. *Classifications*

Game theory is often classified based on different game models. A game theory model serves as a mathematical abstraction of a game scenario, representing the core components: players, strategies, and payoffs. Below are several common game models.

2.4.1. *Zero-sum games and non-zero-sum games*

(1) Zero-sum game

Zero-sum games are characterized by a situation where one player's gain is exactly balanced by the other player's loss, resulting in total gains that sum to zero. A fundamental concept in analyzing such games is the minimax theorem, which can be expressed as:

$$\max_{\sigma_1}\min_{\sigma_2} U_1(\sigma_1,\sigma_2) = \min_{\sigma_2}\max_{\sigma_1} U_1(\sigma_1,\sigma_2) \tag{1}$$

where $U_1(\sigma_1,\sigma_2)$ are the payoff for player 1 when strategies σ_1 and σ_2 are employed by players 1 and 2, respectively.

(2) Non-zero-sum game

Non-zero-sum games are characterized by situations where the total gains or losses of all players are not fixed and may vary depending on strategic choices. Unlike zero-sum games, these interactions allow for outcomes where all players can benefit (positive-sum) or suffer losses (negative-sum), enabling both cooperative and competitive dynamics.

2.4.2. *Cooperative games and non-cooperative games*

(1) Cooperative game

In cooperative games, players can form coalitions and allocate total gains through binding agreements. The core concepts involve efficiency (i.e., Pareto optimality), dummy players, symmetry, and additivity.

Pareto optimality refers to a distribution where no player can be made better off without worsening the outcome for at least one other player. Mathematically, this requires that the total payoff allocated to all players equals the maximum value of the grand coalition.

The Shapley value is an important solution concept in cooperative game theory. It calculates each player's average contribution by considering all possible orders of coalition formation [27]:

$$\phi_i = \frac{1}{n!}\sum_{\substack{S\subseteq N\\ i\notin S}}\ \sum_{\pi\in\Pi(S\cup\{i\})} (v(S\cup\{i\}) - v(S)) \tag{2}$$

where n is the total number of players in the game; $N = \{1,2,\dots,n\}$ represents the set of all players; $S \subseteq N$ denotes a subset of players (a coalition); $i \notin S$ means player i is not in coalition S; $\Pi(S)$ represents the set of all possible permutations (orderings) of players in coalition S; $v : 2^N \rightarrow \mathbb{R}$ is the characteristic function that assigns a value to each coalition, representing the total payoff that the coalition can guarantee for itself (where 2^N denotes the power set of N, i.e., all possible subsets of N); $\Delta v(i,S) = v(S\cup\{i\}) - v(S)$ represents player i's marginal contribution to coalition S; ϕ_i is the Shapley value of player i. Notably, the Shapley value satisfies Pareto optimality, as $\sum_{i\in N}\phi_i = v(N)$, ensuring no cooperative surplus is wasted.

(2) Non-cooperative game

In non-cooperative games, players act independently and cannot form binding agreements. A non-cooperative game consists of three main elements: a set of players N, each player's strategy set Σ_i, from which player i chooses a strategy σ_i, forming a strategy profile $\sigma = (\sigma_1,\sigma_2,\dots,\sigma_n)$, and a payoff function U, which assigns a payoff $U_i(\sigma)$ to each player i based on the chosen strategies. A key concept in analyzing these games is the Nash equilibrium, which occurs when each player's strategy is optimal, given the strategies of all other players. Mathematically, this can be expressed as:

$$U_i(\sigma_1^*,\sigma_2^*,\dots,\sigma_i^*,\dots,\sigma_n^*) \geq U_i(\sigma_1^*,\sigma_2^*,\dots,\sigma_i',\dots,\sigma_n^*) \tag{3}$$

for all possible strategies σ_i' of player i, where U_i is the payoff for player i, and σ_i^* denotes the equilibrium strategy of player i. This means that no player can benefit by changing their strategy while the others keep theirs unchanged. The Nash equilibrium is a fundamental solution concept in non-cooperative game theory, representing a state where no player can improve their payoff by unilaterally changing their strategy, given the strategies of the other players.

2.4.3. Complete information games and incomplete information games

(1) Complete information game

In complete information games, all players possess full knowledge of the game's structure, including the available strategies and the payoff functions for every player. These games can be static, where players make decisions simultaneously without observing the choices of others, or dynamic, involving sequential moves. However, in complete information static games, there is no private information, and players act based on common knowledge of the game's parameters.

(2) Incomplete information game

In incomplete information games, players lack complete knowledge about certain aspects of the game, such as other players' payoff functions or strategies [30]. These games are characterized by information sets that encompass multiple decision nodes, meaning players are uncertain about the specific path of the game when they reach a particular decision point. In Bayesian games, this uncertainty is modeled by assigning types to players, where each type represents a possible configuration of a player's private information. A Bayesian game can be formally represented by the tuple G:

$$G = (N, (T_i)_{i\in N}, (\Sigma_i)_{i\in N}, (U_i)_{i\in N}, P) \tag{4}$$

where T_i denotes the type space for player i; P is the probability distribution over the types of players, reflecting the uncertainty each player has about others' types.

2.4.4. Static game and dynamic game

(1) Static game

Static games are characterized by simultaneous decision-making among players, where each player formulates strategies without prior knowledge of others' choices. This class of games is commonly formalized through a payoff matrix, a tabular representation that delineates the outcomes associated with every possible combination of players' strategies. Each cell within the matrix encapsulates the payoffs received by all players for a specific strategic profile, providing a comprehensive overview of the game's potential results.

(2) Dynamic game

Dynamic games, in contrast, unfold over time with sequential decision-making processes, compelling players to incorporate the game's historical progression into their strategic calculus [46]. These games are typically modeled using either the extensive form, which graphically represents the sequence of moves, or recursive structures. The Bellman equation serves as a cornerstone in solving dynamic programming problems within the context of dynamic games, expressed as:

$$V(s) = \max_{\alpha}\{u(s,\alpha) + \beta \sum_{s'} P(s'|s,\alpha)V(s')\} \tag{5}$$

where $V(s)$ denotes the value function, signifying the expected payoff at a given state s. The term $u(s,\alpha)$ represents the immediate payoff obtained from taking action a in state s, β is the discount factor that accounts for the time-value of future payoffs, and $P(s'|s,\alpha)$ is the transition probability of moving from state s to state s' upon executing action α.

Notably, differential games, an integral and highly significant subclass of dynamic games, exhibit a profound connection with optimal control theory. Differential games are designed to model scenarios where strategic interactions occur continuously over time, and the state evolution of the game is governed by systems of differential equations. The state dynamics in differential games are typically articulated through a system of ordinary differential equations:

$$\dot{\mathbf{s}}(t) = \mathbf{f}(\mathbf{s}(t), \mathbf{v}_1(t), \mathbf{v}_2(t), \dots, \mathbf{v}_n(t), t) \tag{6}$$

where $\mathbf{s}(t)$ represents the state vector, encapsulating the complete state of the game at time t, $\mathbf{v}_i(t)$ are the control strategies of the ith player, continuously adjusted over time, and $\mathbf{f}$ prescribes the state evolution based on the current state, players' control inputs and time.

Players in differential games generally aim to optimize a cost or payoff functional, often expressed as an integral over time:

$$J_i = \int_{t_0}^{t_f} L_i(\mathbf{s}(t), \mathbf{v}_1(t), \dots, \mathbf{v}_n(t), t)dt + \Phi_i(\mathbf{s}(t_f)) \tag{7}$$

where J_i is the payoff functional for player i, L_i is the running cost function, Φ_i is the terminal payoff function, t_0 and t_f are the initial and final times, respectively.

Solving differential games relies on advanced mathematical tools, with the Hamilton–Jacobi–Bellman (HJB) equations playing a crucial role. For a single-player differential game, the HJB equation is given by:

$$-\frac{\partial V}{\partial t} = \min_{\mathbf{v}} \left\{ H(\mathbf{s}, \frac{\partial V}{\partial \mathbf{s}}, \mathbf{v}, t) \right\} \tag{8}$$

where $H(\mathbf{s}, \frac{\partial V}{\partial \mathbf{s}}, \mathbf{v}, t) = \frac{\partial V}{\partial \mathbf{s}}^T \mathbf{f}(\mathbf{s}, \mathbf{v}, t) + L(\mathbf{s}, \mathbf{v}, t)$ is the Hamiltonian function, and $V(\mathbf{x}, t)$ is the value function. In multi-player scenarios, the HJB equations are extended to account for the strategic interactions among players, providing a theoretical framework to derive optimal strategies in continuous-time dynamic games.

2.4.5. Stochastic games and evolutionary games

(1) Stochastic game

Stochastic games incorporate randomness, with payoffs determined by players' strategies and random variables [47]. A simple representation of a random game can be given by

$$EU = \sum_{\omega\in\Omega} P(\omega)\cdot U_i(\sigma,\omega) \tag{9}$$

where EU is the expected utility; $P(\omega)$ is the probability of state ω; $U_i(\sigma,\omega)$ is the payoff for player i in state ω given strategy profile σ.

(2) Evolutionary game

Evolutionary games model the evolutionary dynamics in populations where individuals interact, with fitness depending on their strategies [46]. A fundamental model for strategy evolution is the replicator dynamics, which can be expressed as

$$\dot{p}_i = p_i \left(U_i(\sigma) - U(\sigma)\right) \tag{10}$$

where $\dot{p}_i$ is the rate of change of the frequency of strategy i, $U_i(\sigma)$ is the average payoff of players using strategy i, and $U(\sigma)$ is the mean payoff of the population.

2.5. Conclusion

In summary, game theory is extensively applied across various technological domains, particularly in computer science, electronics, and aerospace engineering, where it supports strategic decision-making and problem-solving [48–52]. It offers a comprehensive framework for understanding strategic interactions among rational decision-makers. By analyzing the fundamental elements, temporal and spatial contexts, cognitive rationality, and organizational structures, we gain insights into the complexities of decision-making processes and their resulting outcomes.

Table 2
Evolution and characteristics of MASs.

Period	Contributor(s)	Contributions	Characteristics
1970–1989	DeGroot	Statistical consensus theory [55]: https://doi.org/10.1080/01621459.1974.10480159	Consensus through communication
	Lamport	Event synchronization in distributed systems [56]: https://doi.org/10.1145/359545.359563	Mechanisms for consistency and reliability
	Smith	Contract net protocol [57]: https://doi.org/10.1109/TC.1980.1675516	Distributed task allocation via competition and negotiation
	Bratman	Belief–desire–intention model [58]: https://doi.org/10.4159/harvard.9780674984556	Description and prediction of individual actions
	Minsky	The society of mind [59]: https://doi.org/10.5555/22939	Complex behavior through simple unit interactions
	Reynolds	Distributed behavioral model [60]: https://doi.org/10.1145/37402.37406	Individual behavior in distributed systems
1990–2009	Vicsek	Discrete-time model [61]: https://doi.org/10.1103/PhysRevLett.75.1226	Study of crowd behavior (e.g., bird flight, fish swimming)
	Fax	Optimal and cooperative control [62]: https://doi.org/10.1109/TAC.2004.834433	Emphasis on coordination among individuals
	Tanner et al.	Consensus under switching topology [63]: https://doi.org/10.1109/CDC.2003.1272911	Group consensus in dynamic networks with changing topology
	Olfati	Average consensus under undirected and balanced graphs [64]: https://doi.org/10.1109/TAC.2004.834113	Maintaining network stability and fairness
2010–Present	Wen et al.	Sequence modeling multi-agent reinforcement learning (MARL)[65]: https://doi.org/10.48550/arXiv.2206.00932	Role of time series data in decision-making
	Bansal	Multi-agent hierarchical reinforcement learning strategy [66]: https://doi.org/10.1126/science.aau6249	Enhancing efficiency and effectiveness

3. Overview of MASs

3.1. Concept

Inspired by natural collaborative systems such as bee colonies, MASs leverage the power of cooperative behavior, where individuals work together to accomplish tasks beyond the capability of a single entity [53]. MASs are distributed computing frameworks composed of autonomous and intelligent agents that interact and collaborate to achieve common objectives. Each agent in a MAS can perceive its environment, make decisions, and communicate with other agents. MASs have broad applications in areas such as robotic systems, distributed decision-making, traffic control, and business management [54].

3.2. Development

In the field of MASs, researchers often focus on agent interactions and coordination to achieve common goals or address complex challenges. This subsection explores the evolution of MASs, as detailed in Table 2.

DeGroot's statistical consensus theory elucidates the process by which group members achieve agreement through communication [55], mirroring the equilibrium in game theory where individual strategy choices align with group behavior. Lamport's research on event synchronization in distributed systems provides a foundation for understanding coordination challenges in MASs [56], akin to strategic interactions in game theory. Smith's contract net protocol models market mechanisms for task allocation [57], reflecting concepts of market equilibrium and strategic decision-making in game theory. Bratman's belief–desire–intention model offers a framework for understanding individual actions [58], akin to modeling rational behavior in game-theoretic scenarios. Minsky's *Society of Mind* [59], which explores interactions among simple agents to comprehend complex behaviors, shares similarities with MASs in game theory. Reynolds's distributed behavior model simulates group dynamics [60], offering insights into the dynamic equilibrium of strategies within groups, much like in game theory.

The evolution of research in MASs has been driven by insights from both computational models and biological systems. A milestone is the Vicsek model [61], which describes swarm alignment dynamics through discrete-time updates :

$$\theta_i(t+\Delta t) = \frac{1}{|\mathcal{N}_i|+1}\left(\theta_i(t) + \sum_{j\in\mathcal{N}_i}\theta_j(t)\right) + \eta_i(t) \tag{11}$$

$$\boldsymbol{p}_i(t+\Delta t) = \boldsymbol{p}_i(t) + v_{\mathrm{vel}}\Delta t\begin{bmatrix}\cos\theta_i(t)\\ \sin\theta_i(t)\end{bmatrix} \tag{12}$$

where $\mathcal{N}_i$ denotes the set of neighboring agents of agent i; $\theta_i(t)$ is the heading angle, $\boldsymbol{p}_i(t)$ is the position, v_{vel} is velocity, and $\eta_i(t)$ is noise. This model demonstrates how simple local rules can generate emergent global coordination, inspiring MASs design principles.

Fax's research on optimal and cooperative control highlights the significance of seeking optimal solutions and fostering cooperation within MASs [62], aligning closely with principles in cooperative game theory and strategic optimization. Jadbabaie's work on consensus under switching topologies explores how changes in network structure affect the achievement of consensus [63], paralleling the influence of network topology on strategic interactions in game theory. Olfati's studies on average consensus in undirected and balanced graphs provide theoretical insights into equilibrium states within MAS [64], resonating with the concept of equilibrium in game theory. Additionally, Wen et al.'s exploration of sequential modeling and MARL [65], alongside Bansal's investigations into multi-agent hierarchical reinforcement learning [66], delves into strategy learning and optimization in dynamic environments, closely mirroring the dynamics of games and the learning of equilibria in game theory.

Fig. 2. Elements of MAS.

3.3. Elements

MAS comprise multiple autonomous agents that interact through defined mechanisms within diverse environments to solve complex problems or execute tasks. The fundamental elements of MAS are depicted in Fig. 2.

3.3.1. Agents

(1) State. The state represents the agent's internal representation of the environment, comprising attributes such as position, velocity, health status, and resource quantities, serving as the basis for decision-making and action selection [67]. The dynamic evolution of an agent's state can be modeled as:

$$\dot{x}_i(t) = f\left(x_i(t), v_i(t)\right) + \sum_{j\in\mathcal{N}_i} w_{ij}\left(x_j(t) - x_i(t)\right) \tag{13}$$

where $x_i(t) \in \mathbb{R}^n$ is the state vector, $v_i(t)$ is the control input, $\mathcal{N}_i$ is the neighbor set, and w_{ij} is the interaction weight. Here, the function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ represents the agent's internal dynamics, which could encode state transitions, parameterized models, or state-dependent target functions. For instance, f might describe physical laws governing the agent's motion, learned behavioral policies, or adaptive decision-making rules under dynamic contracts.

(2) Action. Actions are the behaviors performed by agents to alter their state or influence the environment, selected based on their current state and objectives, and can be predefined or learned.

(3) Perception. Perception involves agents acquiring information about their state and environment, either directly through sensors or indirectly via communication [68].

(4) Communication. Communication enables agents to exchange information, facilitating coordination of actions and sharing of knowledge [69].

(5) Goal. Goals are the objectives that agents aim to achieve, guiding their action selection to align with these goals [70].

(6) Learning. Learning allows agents to improve their behavior to better achieve their goals, incorporating methods such as supervised, unsupervised, or reinforcement learning [71,72].

(7) Adaptability. Adaptability refers to an agent's capacity to adjust its behavior in response to environmental changes, enabling it to address new challenges [73].

(8) Interaction. Interaction involves agents engaging with both the environment and other agents, influencing each other's states and behaviors [74].

3.3.2. Environment

The environment in MAS defines the action space and interaction modes of the agents, consisting of two main aspects.

(1) Physical environment. The physical environment is the tangible or simulated space where MAS operates, involving robots, drones, or automated devices in real-world settings, or software agents in virtual spaces [75]. It includes spatial dimensions and physical properties such as obstacles, terrain, and weather conditions, which influence agent behavior and decision-making [76]. The state of the physical environment $s_p(t) \in \mathcal{S}_p$ can be modeled as a Markov decision process:

$$s_p(t+1) \sim P\left(s_p(t+1) \mid s_p(t), \{\alpha_i(t)\}_{i=1}^N\right) \tag{14}$$

where $P(\cdot)$ is the transition probability dependent on agent actions.
Agents must perceive the environment, plan paths, avoid obstacles, and interact physically with other agents or objects.

(2) Information environment. The information environment facilitates agent communication and information sharing, encompassing communication networks, data-sharing platforms, and sensor networks [77]. Its core feature is the ability to process and exchange information, requiring agents to efficiently receive, process, and transmit data [78].
Communication channels are constrained by bandwidth B and latency τ, leading to delayed information exchange:

$$\hat{x}_j(t) = x_j(t-\tau) + \epsilon(t) \tag{15}$$

where $\epsilon(t) \sim \mathcal{N}(0, \sigma^2)$ models Gaussian noise.
The quality of this environment directly impacts agents' decision-making and coordination. An effective information environment enhances communication, improving system efficiency and responsiveness, while limitations like delays or security issues can hinder performance [79].

3.3.3. Interaction mechanism

(1) Communication. Information exchange between agents is essential for collaboration and coordination. It can be synchronous or asynchronous, point-to-point or broadcast, and relies on well-designed protocols and languages to ensure accurate transmission and comprehension of information [80].

(2) Collaboration. Agents achieve shared objectives through collaboration, involving task allocation, resource sharing, and result integration. Effective collaboration relies on trust and coordination mechanisms to ensure each agent contributes meaningfully to the collective goal [81].

(3) Competition. In resource-constrained scenarios, agents may compete for resources to fulfill their objectives. Competition mechanisms must be designed to prevent excessive resource consumption and maintain system stability [82].

(4) Learning and adaptation. Agents enhance their behavior and strategies through learning, enabling better adaptation to environmental and task demands. Learning can occur individually or collectively, with agents leveraging both their own experiences and those of others [83].

(5) Coordination. Coordination ensures consistency in agent actions by synchronizing decisions and resolving conflicts. A canonical protocol is:

$$v_i(t) = -\sum_{j\in\mathcal{N}_i} a_{ij}\left(x_i(t) - x_j(t)\right) \tag{16}$$

which guarantees asymptotic convergence to the average consensus:

$$\lim_{t\to\infty} x_i(t) = \frac{1}{N}\sum_{j=1}^{N} x_j(0) \tag{17}$$

This consensus state represents a stable point in the dynamical system rather than a game-theoretic equilibrium; it relates to game theory only when explicitly reformulated as a potential game. Effective coordination mechanisms prevent conflicts and redundant efforts, enhancing overall system efficiency [84].

(6) Negotiation. In complex MAS, agents negotiate to reach agreements and resolve resource or task allocation conflicts [85]. Negotiation mechanisms balance competition and collaboration, fostering mutually beneficial outcomes.

(7) Social structure. Agents may establish social hierarchies, such as leader–follower relationships, influencing interaction and decision-making dynamics [86]. These structures form the intricate interaction network of MAS, enhancing flexibility and adaptability. Effective system operation requires careful integration of these elements to ensure efficiency and stability.

These elements constitute the core of MAS, allowing them to address challenges in complex environments through agent interaction and coordination.

3.4. Classifications

Graph theory, a fundamental branch of mathematics, provides a rigorous framework for analyzing the properties of graphs – mathematical structures that model pairwise relationships between objects [87]. In the context of MASs, this formalism offers profound insights into system architectures. Agents are represented as nodes (vertices) in a graph $G = (V, E)$, where $V = \{1, 2, \dots, N\}$ denotes the agent set, and edges $E \subseteq V \times V$ encode communication or interaction patterns.

3.4.1. Graph-theoretic foundations

The adjacency matrix $A = [a_{ij}]$ and Laplacian matrix L are two fundamental descriptors of graph connectivity. The adjacency matrix entries are defined as:

$$a_{ij} = \begin{cases} 1 & \text{if } (j, i) \in E \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

while the Laplacian matrix $L = [l_{ij}]$ captures both connectivity and degree information:

$$l_{ij} = \begin{cases} -a_{ij} & \text{if } i \neq j \\ \sum_{k \neq i} a_{ik} & \text{if } i = j \end{cases} \tag{19}$$

Equivalently, $L = D - A$, where D is the diagonal degree matrix with $D_{ii} = \sum_j a_{ij}$. The eigenvalues of L, particularly the second smallest eigenvalue $\lambda_2(L)$ (algebraic connectivity), govern consensus convergence rates in MASs.

These mathematical tools enable systematic classification of MASs based on topological properties [88,89], with four primary types identified as shown in Fig. 3.

3.4.2. Centralized MAS

Centralized architectures adopt star topologies (Fig. 3a), where a central coordinator node $c \in V$ connects to all peripheral agents [3]. The adjacency matrix $A = [a_{ij}]$ satisfies:

$$a_{ij} = \begin{cases} 1 & \text{if } j = c \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

This architecture is particularly effective in scenarios requiring a single point of control, such as centralized task allocation or coordinated decision-making. The central node orchestrates the actions of all agents, ensuring efficient and coherent system behavior. The primary advantage of this structure lies in its simplicity and clear hierarchical command chain, which facilitates rapid decision-making [90]. However, the system is inherently vulnerable to single-point failures: the failure of the central node can lead to a complete system breakdown [91]. Additionally, scalability is limited, as the central node becomes a bottleneck with increasing numbers of agents.

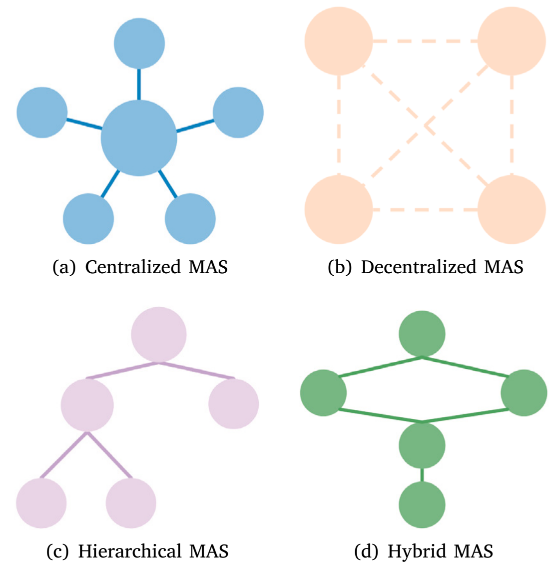

Fig. 3. Classifications of MAS.

3.4.3. Decentralized MAS

Decentralized systems utilize complete graphs (Fig. 3b), where each agent directly interacts with all others ($N_i = V \setminus \{i\}$) [89]. The Laplacian matrix $L = D - A$ (see Eq. (19)) exhibits full connectivity through its eigenvalues:

Theorem 1. *For a decentralized multi-agent system with an undirected communication graph, consensus is achievable if and only if the algebraic connectivity satisfies* $\lambda_2(L) > 0$.

This architecture is ideal for applications requiring high levels of autonomy and robustness, such as blockchain networks or distributed sensor systems. In decentralized MAS, agents operate independently, making decisions based on local information and interactions with neighboring agents. The absence of a central coordinator eliminates single points of failure, enhancing system resilience. However, the high degree of connectivity results in significant communication overhead, and the lack of centralized control can lead to coordination challenges, particularly in large-scale systems [9,92].

3.4.4. Hierarchical MAS

Hierarchical architectures employ tree graphs (Fig. 3c), where nodes are organized into multiple levels of authority with layered communication [93]. Let L_{tree} be the Laplacian matrix, its block-diagonal structure:

$$L_{\text{tree}} = \bigoplus_{k=1}^{K} L_k \tag{21}$$

reflects parent–child relationships between supervisory agents (higher-level nodes) and operational agents (lower-level nodes). Here, $\bigoplus$ denotes the direct sum of submatrices, and each block L_k corresponds to a subsystem in layer k.

Higher-level nodes make strategic decisions, while lower-level nodes carry out specific tasks. This architecture is well-suited for large-scale systems that require a balance between centralized control and distributed execution, such as military command structures or organizational management systems. The hierarchical structure reduces communication overhead and enhances decision-making efficiency by localizing interactions within levels [94]. However, the system may suffer from bottlenecks at higher levels, and its rigid structure may limit adaptability in dynamic environments [95].

3.4.5. Hybrid MAS

Hybrid systems (Fig. 3d) combine multiple topologies. For instance, local clusters may form complete subgraphs (A_{cluster}) while a coordinator manages inter-cluster links (A_{global}), yielding composite adjacency matrices:

$$A = A_{\text{global}} + \bigoplus_{k=1}^{K} A_{\text{cluster}_k} \tag{22}$$

where A_{cluster_k} represents the adjacency matrix of internal connections within the kth cluster. This architecture balances global coordination and local autonomy.

Hybrid systems combine elements of centralized, decentralized, and hierarchical structures, forming composite graphs that leverage the strengths of each approach. For example, a hybrid MAS might feature a central coordinator for high-level decision-making while allowing local clusters of agents to operate autonomously. This architecture is particularly advantageous in complex systems that require both global coordination and local adaptability, such as smart grids or mixed-initiative robotic teams [96]. The hybrid structure offers flexibility and scalability, enabling the system to handle diverse and dynamic tasks [97]. However, designing and implementing hybrid MAS is inherently complex, as it requires careful integration of different control paradigms and may involve trade-offs between centralized efficiency and decentralized robustness [98].

3.5. Conclusion

In summary, MASs enable autonomous agents to collaborate, communicate, and solve complex problems in distributed environments. Its core elements-agents, environments, and interaction mechanisms-ensure adaptability and coordination, while its classifications offer tailored solutions for diverse challenges. Through the content discussed above, we can better analysis and explore the application of MASs.

4. Analysis of the relationships between game theory and MASs

4.1. Conceptual boundaries of related frameworks

This review adopts a precise methodological criterion for classifying works as game-theoretic: strategic interdependence and equilibrium concepts must form the core analytical framework. Strategic interdependence is defined as the condition where agents' payoffs explicitly depend on the actions of other agents. Based on this criterion, we clarify distinctions among six paradigms frequently encountered in aerospace multi-agent systems research:

(1) Pure game theory as a modeling framework for strategic interaction. Pure game theory provides the foundational framework for analyzing scenarios where agents' payoffs explicitly depend on others' actions. Equilibrium concepts such as Nash equilibrium for non-cooperative settings and Shapley value for cooperative allocations serve as its core analytical tools, enabling rigorous analysis of competitive and collaborative interactions among autonomous aerospace platforms [26,27].

(2) Mechanism design as the inverse problem of game theory. Mechanism design constitutes the inverse problem of game theory: rather than predicting outcomes given fixed rules, it designs incentive-compatible rules to achieve desired system-level objectives. In aerospace applications, this paradigm manifests in spectrum auction mechanisms for satellite networks [99] and task allocation protocols where agents truthfully report capabilities without strategic manipulation.

(3) Distributed optimization as coordination without strategic modeling. Distributed optimization seeks global optima through local computations without modeling strategic incentives among agents. Consensus algorithms for formation control [64] and distributed gradient descent exemplify this paradigm. Such methods relate to game theory only when explicitly reformulated as potential games with well-defined potential functions satisfying marginal contribution equivalence; otherwise they represent coordination without strategic interdependence.

(4) MARL as an adaptive solution method. MARL is an adaptive learning framework where agents improve policies through environmental interaction. While MARL algorithms may target game-theoretic equilibria as learning objectives, MARL itself is distinct from game theory: the former serves as a solution method for decision-making under uncertainty, while the latter provides a modeling framework for strategic interactions.

(5) Evolutionary dynamics as population-level strategy evolution. Evolutionary dynamics describes strategy frequency evolution in populations through differential equations such as replicator dynamics, without assuming individual rationality or equilibrium reasoning [46]. This contrasts sharply with classical game theory's assumption of rational equilibrium-seeking behavior, making evolutionary models suitable for bio-inspired swarm behaviors but conceptually distinct from strategic game analysis.

(6) Mean-field approximation as a computational scaling technique. Mean-field approximation provides a tractable computational framework for ultra-large-scale MASs by assuming negligible individual impact on population distribution [100]. It serves as a scaling technique rather than a distinct game-theoretic paradigm; mean-field games approximate N-player differential games in the limit N approaching infinity, but lose fidelity for small-to-medium agent populations where individual interactions significantly influence system behavior.

These distinctions carry practical implications for aerospace system design: satellite formation control typically employs distributed optimization without strategic modeling; spectrum sharing requires mechanism design with explicit incentive alignment; and swarm-vs-swarm engagement demands pure game-theoretic analysis of adversarial incentives. Misclassifying these paradigms risks inappropriate method selection and flawed system architecture decisions.

4.2. Mapping analysis

Game theory and MASs are closely related yet distinct fields. Game theory provides a framework for analyzing and predicting individual behaviors and strategies in interactive scenarios, while MASs focus on designing and implementing autonomous agents capable of interacting with one another. In this paper, we primarily summarize the key mapping relationships between the elements of game theory and MASs as follows.

(1) Players and agents.
In game theory, players are rational decision-makers aiming to maximize their payoffs [85]. In MASs, these players are instantiated as agents, autonomous entities that interact within an environment to achieve their goals. Each agent in a MAS can be regarded as a player in game theory, selecting action strategies based on their own objectives and interacting or competing with other agents. In aerospace applications, the game-theoretic "players" correspond to autonomous agents such as multi-unmanned aerial vehicle (UAV) systems, satellite constellation networks, or orbital spacecraft. These agents function as independent decision-making units that perform diverse operational tasks. Within satellite constellation mission planning, each satellite operates as a self-optimizing agent whose behavior can be modeled as strategic choices within a game-theoretic framework. The decision-making processes of these agents are driven not only by individual objectives but also by the necessity

to account for collaborative, competitive, and interdependent relationships with other agents. This dynamic becomes particularly evident in spectrum resource allocation and orbital window scheduling scenarios, where multiple satellites within a constellation must coordinate to maximize collective benefits while preventing conflicts arising from self-interested behaviors [101].

(2) Strategies and actions.

Strategies in game theory represent the plans of action available to players. In MASs, these correspond to the actions agents can execute, typically influenced by their current state and objectives. In aerospace mission execution, strategies and actions hold paramount importance. Unmanned aerial vehicles operating in hostile environments must dynamically adapt their flight paths, a process that can be optimized through game-theoretic strategies. When mission objectives such as reconnaissance or strike operations demand flexibility, individual UAVs can adjust their tactical approaches in real-time, while formation-based agents coordinate to accomplish complex tasks through collaborative strategies. Furthermore, dynamic airspace resource allocation scenarios — where multiple aircraft must resolve conflicts over shared safety corridors — can be effectively modeled using cooperative game theory. This framework enables the minimization of conflicts and resource wastage within shared airspace, ensuring optimal utilization of limited aerial resources while maintaining operational safety and efficiency [102,103].

(3) Payoffs and goals.

Payoffs in game theory represent the outcomes or rewards that players receive based on the strategies chosen [104]. In MASs, these are analogous to goals, which guide agents in selecting actions that align with their objectives. The payoff concept in game theory directly maps to multi-objective optimization problems encountered in aerospace scenarios. Typical payoff objectives include energy consumption minimization and mission data acquisition efficiency. Effective payoff modeling enables the optimization of aerial combat maneuvering decisions under dynamic conditions, thereby enhancing mission completion quality. Furthermore, the collaborative observation capabilities among constellation satellites can be significantly improved through payoff models that maximize collective sensing performance while balancing individual satellite constraints [103,105].

(4) Information.

Game theory categorizes games into two classes: games of complete information and games of incomplete information. In MASs, this classification corresponds to the perception and communication capabilities of agents—agents may possess either complete knowledge or partial information regarding the environment and other agents' states [106].

In aerospace systems, reconnaissance drones frequently encounter partial observability of adversary movements, where adversaries may employ deceptive tactics such as electronic interference or decoy deployment, creating significant uncertainty in target states. Agents must dynamically evaluate information credibility and balance operational actions with reconnaissance accuracy during missions. Such information uncertainty can be effectively addressed through partially observable stochastic games frameworks [107].

Within large-scale autonomous swarm formations, communication delays and partial perception capabilities further necessitate game-theoretic approaches for decision modeling. In satellite constellation scenarios, different nodes may observe target states with spatio-temporal inconsistency, leading to partial observability challenges. Data fusion, error reduction, and perception quality optimization become critical issues. Public goods game theory can facilitate distributed trust evolution collaboration models to enhance information fusion performance of unmanned swarms in competitive scenarios.

In time-critical missions such as asteroid defense or high-speed target interception, systems must identify optimal trade-offs between observation duration and action timing. Within finite time windows: extended observation periods may result in missed operational windows, while premature actions based on incomplete information could lead to severe decision errors. Sequential game modeling provides a solution by treating information acquisition itself as a strategic variable and identifying the equilibrium between information gathering and action execution through sequential decision processes.

These dynamic frameworks enable agents to select risk-minimizing paths and strategies under information constraints, thereby enhancing mission execution efficiency and system robustness while maintaining operational effectiveness in complex aerospace environments [108].

(5) Environment.

The environment in game theory can range from deterministic to stochastic. In MASs, this is mirrored in the physical environment and information environment, which define the action space and interaction modes of agents [109]. Environmental factors further amplify the complexity of game-theoretic problems in aerospace applications. Deep space exploration missions exemplify this complexity, where spacecraft must operate within hybrid environments that integrate deterministic elements such as orbital dynamics with stochastic phenomena including solar storms, electromagnetic interference, and space debris collision risks. Simultaneously, the challenge of unmanned aerial vehicles dynamically adjusting their flight trajectories in turbulent wind fields represents a canonical stochastic dynamic game scenario. Agents operating within such environments must not only optimize individual strategies but also rapidly adapt to external perturbations while balancing the dual influences of deterministic predictability and stochastic uncertainty [103]. This capability ultimately enhances system resilience and operational effectiveness across complex aerospace domains.

(6) Logic, rationality, and intelligence.

Game theory provides a formal framework for modeling strategic interactions among rational decision-makers. In MASs, agents exhibit learning and adaptive capabilities under the realistic constraints of bounded rationality and finite reasoning depth [110]. The operational performance of aerospace vehicles remains inherently constrained by computational resource limitations, energy allocation bottlenecks, and real-time communication bandwidth constraints. These physical and operational boundaries align precisely with the bounded rationality framework [111,112], which offers an effective theoretical lens for characterizing system behavior under resource constraints.

Deep-space probes face significant challenges in sustaining long-term operations due to constrained energy supplies, such as the limited lifespan of radioisotope thermoelectric generators or declining solar panel efficiency [113]. These energy limitations prevent continuous execution of high-energy optimal computations, necessitating innovative approaches to autonomous decision-making [114]. Under conditions of extreme energy scarcity, deep-space probes must make difficult trade-offs among scientific data collection, communication, and system maintenance. Consequently, agents adopt "satisficing" strategies rather than optimization approaches, selecting "good enough" solutions rather than optimal ones. This behavioral adaptation leads systems to employ heuristic rules and pre-computed strategy libraries, which fundamentally represent manifestations of rational boundaries in practical aerospace applications.

(7) Timing, dynamic and static.
Game theory differentiates between static and dynamic games. In MASs, this translates to the timing elements of strategic interactions, with recurrent games and hybrid games featuring repeating patterns and a combination of static and dynamic elements, respectively [115]. Time factors, often intertwined with task dynamics and planning, map to specific practical scenarios in aerospace: short-term strategy optimization (static games) involves the short-term dynamic scheduling of multiple aerial vehicles in a designated airspace, such as coordinating the takeoff and landing of UAVs and manned aircraft; long-term path decision-making (dynamic games), by contrast, requires constellation-networked satellites to perform dynamic path optimization based on orbital decay and mission frequency to ensure long-term communication stability [116,117].

(8) Relationships, networks, and systems.
Game theory examines relationships among players, including non-cooperative and cooperative dynamics. In MASs, these are seen in the form of interaction mechanisms such as communication, collaboration, competition, and social structures that affect how agents interact within the system [118]. In aerospace applications, the network topology of agent relationships becomes critically important in determining system-wide performance and resilience. Satellite constellations exemplify hybrid cooperative-competitive networks where individual spacecraft must collaborate for data relay while competing for limited ground station access time and spectral resources. This complex relationship structure can be effectively modeled through bipartite consensus frameworks that account for both cooperative and competitive interactions within the same network topology [119]. Similarly, in distributed air traffic management systems, aircraft agents form dynamic alliance networks where temporary coalitions emerge to optimize airspace utilization while maintaining competitive advantages in routing efficiency. The robustness of these aerospace multi-agent networks is further challenged by denial-of-service attacks and communication failures, requiring game-theoretic approaches that can maintain system stability despite adversarial disruptions to the cooperative-competitive balance [120]. In military aerospace applications, networked combat systems leverage multi-agent game strategies to optimize kill chain effectiveness, where the relationship dynamics between reconnaissance, command, and strike agents must be precisely coordinated to achieve mission objectives under competitive pressure from adversarial forces [121].

(9) Quantity.
The concepts of quantity in game theory, such as the number of players and strategies, can be finite or infinite. In MASs, this is represented by the number of agents and the complexity of their strategy sets, which can greatly influence the system's behavior and the required strategic analysis [122]. Aerospace systems present unique challenges in scalability due to the vast range of agent quantities involved in different applications [123]. Small-scale scenarios, such as formation flying of 3–5 UAVs for reconnaissance missions, can leverage complete information game models where each agent can track all others' strategies. However, large-scale deployments like mega-constellations comprising hundreds or thousands of satellites require fundamentally different game-theoretic approaches that account for population dynamics and mean-field approximations. The computational complexity grows exponentially with agent quantity, necessitating distributed algorithms that can achieve near-optimal solutions without requiring global knowledge [124]. In air traffic management systems handling hundreds of aircraft simultaneously, the quantity problem becomes particularly acute, requiring hierarchical game structures where local interactions are optimized while maintaining global flow efficiency [125]. The scalability challenge is further compounded by heterogeneous agent capabilities within the same system, where different aircraft types or satellite classes possess varying strategy sets and computational resources, requiring adaptive game formulations that can accommodate both finite and infinite strategy spaces within integrated aerospace networks.

(10) Preferences.
Player preferences in game theory can be numerical or ordered, influencing decision-making processes [126]. In MASs, preferences guide agents' decision-making, especially in cooperative and competitive scenarios, and can be represented through various types such as numerical values or ordered relationships. Aerospace MASs exhibit sophisticated preference structures that must account for multiple, often conflicting objectives under stringent safety constraints. In autonomous aerial refueling operations, tanker and receiver aircraft must negotiate preferences that balance fuel efficiency, mission time, safety margins, and operational priorities. These preferences are rarely purely numerical but often involve lexicographic ordering where safety constraints dominate economic considerations. Similarly, in Earth observation satellite constellations, individual spacecraft must prioritize imaging requests based on scientific value, political urgency, weather conditions, and energy constraints, creating complex preference hierarchies that evolve dynamically with mission conditions [127]. The challenge is further amplified when agents possess private preference information that must be revealed strategically during negotiation processes [128]. Advanced preference modeling techniques, including fuzzy logic representations and prospect theory frameworks, have emerged to handle the uncertainty and risk-aversion characteristics inherent in aerospace decision-making [129]. These approaches enable agents to make robust choices even when preference information is incomplete or when multiple stakeholders impose competing preference structures on the same autonomous system.

(11) Equilibrium and stable states.
Equilibrium in game theory and stable states in MASs represent distinct concepts that may coincide under specific conditions [130]. In game theory, equilibrium denotes a strategic state where no agent can improve payoff by unilateral deviation; in MASs, stable states typically refer to dynamical convergence (e.g., consensus) that lacks strategic interdependence unless explicitly modeled as a game. In MASs, agent behaviors and interactions can be viewed as a gaming process, with the system's stable state analogous to the game's equilibrium. At equilibrium, no agent can improve its payoff by unilaterally changing its strategy. Similarly, MASs achieve stability when agent behaviors and interactions reach a balanced state, enabling normal operation and goal attainment. Stability in MASs is facilitated through mechanisms like cooperation, coordination, and competition. Insights from game theory's equilibrium concepts and solution methods provide valuable theoretical support for analyzing and controlling stability in MASs. In aerospace applications, equilibrium concepts manifest in critical operational scenarios where system stability directly impacts mission success and safety. For satellite constellation networks, Nash equilibrium solutions enable stable resource allocation patterns where no satellite can improve its communication throughput by unilaterally changing its transmission strategy, preventing destructive oscillations in network behavior [131]. In distributed air traffic control systems, correlated equilibrium approaches help achieve coordinated flow patterns that minimize conflicts while respecting individual aircraft preferences, creating stable traffic corridors that emerge from decentralized decision-making [132]. The robustness of these equilibria becomes particularly crucial when facing external disturbances such as weather events or

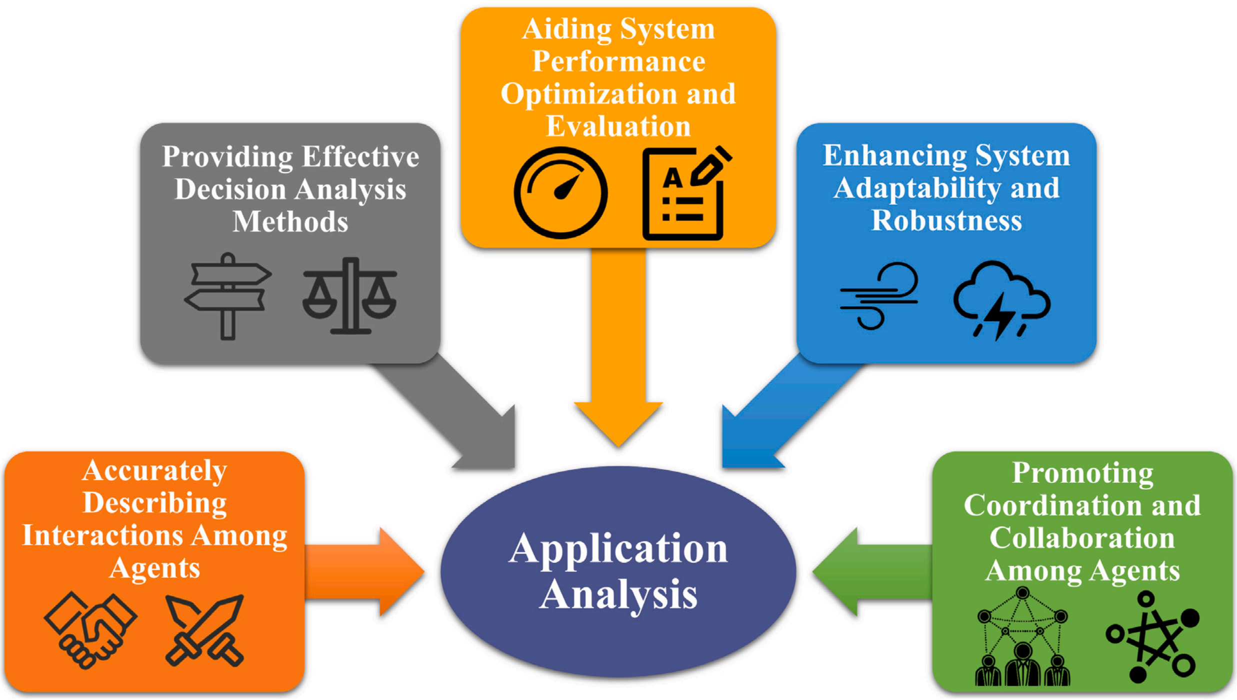

Fig. 4. Application analysis of game theory in MASs.

adversarial jamming, where the system must maintain stability despite perturbations to agent payoffs and strategy spaces. Advanced equilibrium refinement techniques, including trembling-hand perfect equilibrium and subgame perfection, provide the mathematical foundation for designing resilient aerospace MASs that can recover stability after temporary disruptions. These refined equilibrium concepts account for the possibility of off-the-equilibrium play and ensure system stability even when agents make small errors or face unexpected perturbations, ensuring continuous operation in dynamic and uncertain aerospace environments.

These analyses underscore the direct correspondence between players in game theory and agents in MASs, offering a robust framework for understanding and examining the interactions and strategic behaviors within MASs. By leveraging game theory, we can better model and predict agent dynamics, optimize decision-making processes, and enhance system performance in complex environments. This mapping not only bridges theoretical insights but also provides practical tools for designing and managing MASs effectively.

4.3. Application analysis

Game theory provides significant advantages and unique features for analyzing MASs. This subsection delves into the potential applications and synergies between these five fields, as outlined in Fig. 4 and further discussed below.

4.3.1. Accurately describing interactions among agents

(1) Competition relationship modeling. In MASs, agents often have different goals and interests, leading to competition [133]. Game theory precisely describes these competitive relationships by defining agents' strategy spaces and payoff functions, analyzing outcomes under different strategies, and helping researchers understand competitive dynamics.

(2) Cooperation relationship analysis. Beyond competition, agents may also cooperate. Cooperative game theory models and analyzes cooperative behaviors, studying how agents achieve common goals and fairly distribute benefits [134]. For example, in multi-agent cooperative transportation tasks, game theory aids in task and payoff allocation, fostering effective cooperation.

(3) Mixed relationship consideration. In real-world MASs, competition and cooperation often coexist [135]. Game theory comprehensively analyzes these mixed relationships, examining agents' trade-offs and decisions between competition and cooperation, and providing theoretical support for system design and optimization [136,137].

4.3.2. Providing effective decision analysis methods

(1) Strategy selection guidance. Game theory provides a rational decision-making framework for agents, enabling them to choose optimal strategies to maximize payoffs based on others' decisions [138]. By analyzing game models, agents can predict others' behaviors and adjust their strategies accordingly [139]. For instance, in autonomous driving, vehicle interactions can be modeled as a multi-agent game, where each vehicle selects speed and routes based on others' strategies to ensure safety and efficiency [140–142].

(2) Equilibrium state solution. Equilibrium concepts in game theory, such as Nash equilibrium, offer a method to analyze stable states in MASs [143]. At Nash equilibrium, each agent's strategy is the best response to others' strategies, with no incentive to deviate unilaterally. Identifying Nash equilibrium helps determine system stability and agent behavior patterns, which is essential for predicting system dynamics and designing effective control strategies [144].

4.3.3. Aiding system performance optimization and evaluation

(1) System performance optimization. Game theory methods have been demonstrated to enhance MASs performance by modeling strategic interactions among agents [145]. In task allocation problems, game-theoretic frameworks enable the formulation of task distribution as strategic decision-making processes, where optimizing agents' strategies and payoff allocations improves efficiency [146]. Additionally, game-theoretic approaches contribute to system-level optimization through distributed resource

allocation mechanisms, such as dynamic resource distribution and network resource management, thereby indirectly improving system robustness and coordination [147].
(2) System performance evaluation. Game theory offers a quantitative approach to evaluate MASs performances. By defining appropriate payoff functions and evaluation metrics, different system designs can be compared and assessed, enabling the selection of optimal solutions. This supports researchers in scientifically evaluating and enhancing system performance during the design phase [148].

4.3.4. Enhancing system adaptability and robustness

(1) Adapting to dynamic environments. MASs often operate in dynamic environments where agent behaviors and environmental states can change unpredictably. Game theory enables agents to quickly adjust their strategies to adapt to these changes [149,150]. For example, in MASs communications, where network topology and load fluctuate, agents can use game-theoretic methods to optimize communication strategies, ensuring efficiency and reliability [151].
(2) Dealing with uncertainty. MASs face uncertainties such as agent failures and environmental disruptions. Game theory helps agents make rational decisions under uncertainty, improving system robustness [152,153]. For instance, in MASs cooperative combats, agents can adapt their strategies using game-theoretic methods to compensate for failures or adversary interference, ensuring mission success [154].

4.3.5. Promoting coordination and collaboration among agents

(1) Coordination mechanism design. Game theory offers theoretical support for designing coordination mechanisms among agents. By establishing fair game rules and incentive structures [155], agents are encouraged to coordinate and collaborate on tasks effectively [156,157].
(2) Cooperative strategy formulation. Game theory aids agents in developing effective collaboration strategies. By analyzing agents' relationships and interactions, they can identify collaboration opportunities and devise corresponding strategies [158].

These analyses underscore the multifaceted benefits of game theory research in MASs, ranging from precisely modeling agent interactions to improving system adaptability and robustness, fostering effective coordination and collaboration among agents.

5. Application scenarios of game theory in MASs

The integration of game theory and MASs provides a rich theoretical framework for modeling strategic interactions across diverse operational scenarios. This section examines how game-theoretic models are formulated and analyzed within multi-agent contexts, with particular attention to six distinct application domains. Each subsection focuses on specific use cases, exploring the theoretical formulations documented in the literature, emphasizing solution methodologies, and presenting results from experimental implementations and simulation-based validation studies.

5.1. Cooperative control

This subsection examines how game theory has been formulated to address multi-agent coordination challenges, with particular attention to formation control, consensus tracking, and conflict resolution scenarios.

In 2007, Semsar and Khorasani [159] leveraged cooperative game theory to optimize vehicle team collaboration and minimize individual costs by integrating individual cost functions into a unified team cost function and identifying the optimal solution from the Pareto-efficient set using the Nash bargaining approach. Building on this foundation, Aghajani and Doustmohammadi [160] employed cooperative game theory to model the formation control problem of multi-vehicle systems as a differential game, treating each vehicle as an autonomous agent with its individual cost function serving as the payoff function and control input as the strategic decision-making mechanism. Further advancing this field, Khosravi et al. [161] transformed the multi-vehicle cooperative interception of moving targets into a potential game, addressing the complexities of intercepting targets with uncertain arrival times, trajectories, and dynamics under constraints of sensing and communication.

In subsequent developments, Zhao et al. [162] introduced an event-triggered control method for nonlinear MAS with unknown states and disturbances, transforming the optimal consensus tracking problem into a multi-player zero-sum differential game by treating disturbances as control inputs. Jiang et al. [163] proposed a cooperative game theory-based approach for multi-UAV consensus formation control, defining the task as tracking reference states and minimizing team costs to achieve consensus. Tang et al. [164] applied cooperative game theory to develop a model for autonomous aircraft separation assurance, focusing on horizontal cross-conflict scenarios and forming alliances to collaboratively resolve conflicts.

Jiang et al. [165] proposed a cooperative differential game theory-based approach for consensus formation control in multi-UAV systems. They formulated the formation control problem as a dynamic coalitional game with non-transferable utility, aiming to achieve Pareto-optimal consensus by minimizing the weighted team cost function. Tang et al. [166] proposed a game theory-reinforcement learning framework for cooperative decision-making in multi-UAV systems. They formulated the UAV collaboration problem as a public goods game with asymmetric environmental feedback, where cooperators' payoff multipliers dynamically adjust based on global resource constraints. Kusyk et al. [167] presented bio-inspired and game theory based flight control algorithms for autonomous UAV swarms, demonstrating effective coordination in complex airspace environments. Zhou et al. [168] designed an intelligent decision-making and control method for UAV swarm anticollision based on cognitive game theory, providing robust solutions for safety-critical aerospace operations. Jiao et al. [169] introduced a novel control strategy that allows UAVs to autonomously determine optimal flight trajectories without relying on centralized coordination, significantly enhancing system resilience in contested environments.

Liu et al. [170] proposed a game-theoretic decision-making model for multi-vessel collision avoidance, leveraging Bayesian inference and static game theory to predict vessel intentions and formulate real-time decisions under incomplete information by seeking Nash equilibria, enhancing maritime navigation safety and efficiency. Heshami and Kattan [171] introduced a coalitional game theory-based method for cooperative decision-making in self-organizing connected and autonomous vehicles, structured into three key steps as illustrated in Fig. 5. By implementing the aforementioned steps, they framed the lane-changing problem as a dynamic transferable utility issue, aiming to identify Pareto-optimal coalitions for lane-change decisions. This method is designed to improve traffic flow, reduce congestion, and enhance travel time efficiency.

Through the above review of the applications of different types of game theory in various fields of multi-agent cooperative control, it becomes evident that game theory offers a rich variety of ideas and methods for solving multi-agent cooperative control problems. Depending on different application scenarios and control goals, one can flexibly select the appropriate type of game theory and related strategies to achieve efficient, safe, and cooperative control effects. This not only enriches the theoretical framework of multi-agent cooperative control but also provides practical guidance for handling complex real-world situations.

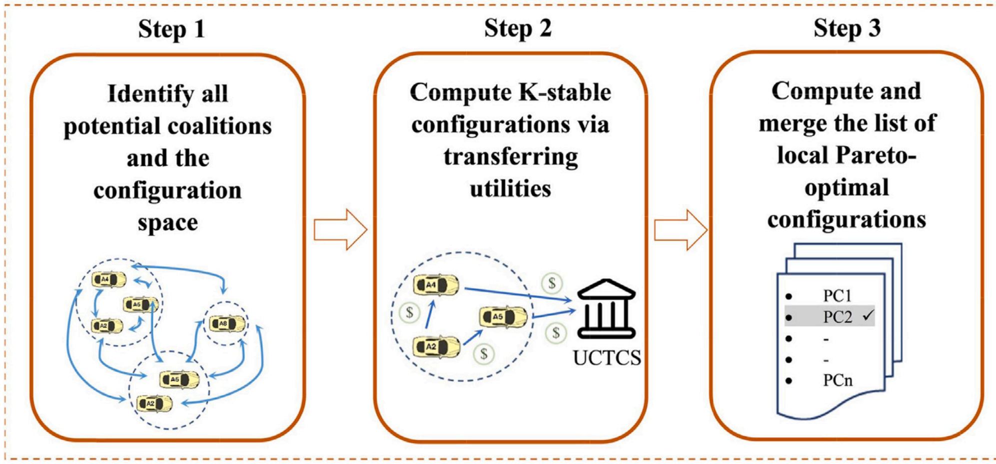

Fig. 5. A three-step model for cooperative lane-changing decisions in self-organizing connected and autonomous vehicles (CAVs) using coalitional game theory. Step 1: Identify potential coalitions and the configuration space. Step 2: Compute K-stable configurations by transferring utilities and allocating them through a universal, collectible, and tradable credit scheme (UCTCS). Step 3: Compute and merge local Pareto-optimal configurations to determine the final lane-changing decisions.
Source: from Ref. [171].

5.2. Cooperative guidance

This subsection explores how game-theoretic approaches have been developed to model strategic interactions in coordinated guidance scenarios, focusing on pursuit-evasion dynamics, swarm navigation, and collision avoidance strategies.

Zhao et al. [172] analyzed the cooperative guidance process for multiple aircraft by formulating flight scenarios using multi-agent Markov game theory and solving them with machine learning algorithms. This approach effectively addressed collision avoidance challenges for a variable number of fixed-wing UAVs in constrained airspace. Chao et al. [173] proposed an integrated guidance strategy combining linear quadratic and norm-bounded game theory, enabling the attacker to evade interception while minimizing fuel consumption and maximizing acceleration for a successful strike. Similarly, Sun et al. [174] modeled the pursuit-evasion scenario of a flight vehicle intercepting an unknown maneuvering target as an evolutionary game process, deriving optimal strategies for both the flight vehicle and the target. These studies highlight the potential of MAS and game theory in enhancing cooperative guidance for complex aerial missions.

Wang et al. [175] established a multi-to-one missile-target interception guidance game model, derived the Nash equilibrium solution using a model prediction algorithm, and developed a cooperative guidance strategy for MASs. Lee et al. [176] implemented singularity avoidance strategies in MAS through cooperative game theory and transfer utility multi-player game theory, effectively resolving inherent singularity issues in pyramid-type control-moment gyro systems. Zhao et al. [177] proposed a stackelberg game-based method to guide heterogeneous robots with one-way communication in MAS for multi-object rearrangement tasks, enhancing efficiency and robustness. Similarly, Zhao et al. [178] formulated the guided trajectory planning problem as a dynamic stackelberg game, enabling a leader robot to strategically direct a follower robot in MAS to collaboratively reach a designated destination. Li et al. [179] developed a game theoretic formulation for multiple UAV cooperative search and surveillance using potential games, achieving efficient area coverage while maintaining formation integrity during extended missions.

Le Ménec employed mean field game theory to model and predict the behavior of large-scale MAS, enabling macro-level collective behavior analysis while maintaining decentralized control at the micro level [100]. This approach facilitated effective swarm guidance in complex environments, particularly in obstacle-rich scenarios, as depicted in Fig. 6. By optimizing the reference trajectory, the swarm achieved congestion anticipation and avoidance, significantly enhancing the overall movement speed of the group. Note that mean-field games serve as a computational approximation for ultra-large-scale MASs by assuming negligible individual impact on population distribution, rather than constituting a distinct game-theoretic paradigm.

The reviewed studies highlight the powerful synergy between game theory and MASs in tackling complex guidance challenges. These approaches significantly enhance the capabilities of autonomous systems, offering robust solutions for collision avoidance, target interception, and collaborative tasks.

5.3. Resource allocation and task distribution

This subsection investigates how game-theoretic frameworks have been applied to address resource allocation challenges in MASs, with emphasis on task distribution, utility optimization, and collaborative decision-making processes.

Lian et al. [180] developed a game-theoretic framework for MAS with linear dynamics, optimizing communication costs by limiting state and control input exchanges while ensuring equitable cost distribution. Similarly, Dong et al. [181] employed hierarchical stackelberg game theory to create a three-tier optimization strategy for microgrid clusters, modeling interactions among cluster agents, microgrid agents, and user agents to enhance collaboration and resource coordination. Cui et al. [182] modeled dynamic resource allocation in multi-UAV networks as a stochastic game, where each UAV's Nash equilibrium strategy optimally responds to others, ensuring no unilateral benefit. They used Markov chains to balance communication costs and system performance. Liu et al. [183] applied MARL, enhanced by game theory, to address resource allocation in IoT networks with edge computing, framing computation offloading as a stochastic game where agents optimize decisions based on local observations. Fu et al. [184] introduced a non-cooperative game-based MARL algorithm for electric vehicle charging scheduling, enhancing station efficiency and minimizing costs.

Li et al. [185] used an aspiration-driven evolutionary game model within the public goods game framework to promote cooperation in unmanned swarms, integrating Lanchester's Law and mean-field game theory to address autonomous collaboration challenges. Chen et al. [186] optimized task selection and spectrum allocation in heterogeneous UAV networks through coalition formation games, treating drones as agents and tasks as strategies to maximize network utility.

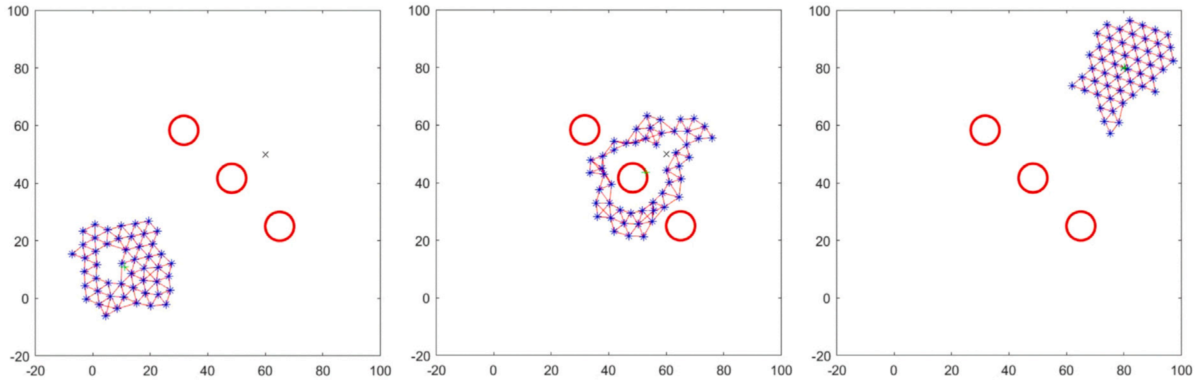

Fig. 6. Swarm guidance of 50 agents marked by blue stars amidst three circular obstacles signified by red circles.
Source: from Ref. [100].

Zhou et al. [187] proposed a game-theoretic method for distributed resource allocation in multiple coalitions, balancing intra-coalition cooperation and inter-coalition conflicts to maximize benefits under resource constraints. Wang et al. [188] developed a dynamic UAV deployment strategy using multi-agent imitation learning to adapt to fluctuating service demands, modeling interactions as a Markov game to analyze Nash equilibrium among UAV operators. Long et al. [189] developed a cooperative game model that effectively allocates tasks and profits among multiple satellites, establishing a robust collaborative mission planning mechanism for Earth observation constellations.

In satellite communication systems, Zhong et al. employed a mechanism design approach by formulating the joint transmit power and bandwidth allocation as a cooperative bargaining framework where rules are designed to achieve incentive-compatible spectrum sharing [99]. As shown in Fig. 7, their approach achieved a Pareto optimal solution that effectively balanced fairness among users while maximizing the total network capacity, demonstrating significant performance improvements over existing methods. This work provides a valuable framework for resource coordination in space-based MASs where spectrum sharing and interference management are critical concerns. Sun et al. [190] formulated the distributed task allocation problem as a potential game to achieve self-organized coordination with minimal communication. Their approach enables satellites to maximize the number of executed observation tasks while minimizing system-level costs through local interactions. As shown in Fig. 8, the task allocation framework involves satellites with overlapping fields of regard that must coordinate to avoid redundant observations while ensuring comprehensive coverage of requested areas.

Research in game theory for MASs task and resource allocation highlight significant progress in improving system efficiency, ensuring fair resource distribution, and promoting agent cooperation. As the field evolves, the integration of game theory with MASs continues to deliver innovative solutions, promising more efficient and intelligent resource allocation. This advancement paves the way for smarter and more adaptive autonomous operations.

5.4. Path coordination and optimization

This subsection reviews how game theory has been utilized to analyze strategic interactions in path planning and coordination scenarios, with particular attention to trajectory optimization, obstacle avoidance, and priority conflict resolution in multi-agent navigation.

Gao et al. [191] proposed a method combining multi-agent deep deterministic policy gradient and game theory to optimize data offloading paths with UAV assistance. They used potential games for service allocation between UAVs and ground users, refining trajectories to balance data offloading delays, energy efficiency, and obstacle avoidance. Similarly, Wang et al. [192] applied game theory to study CAV

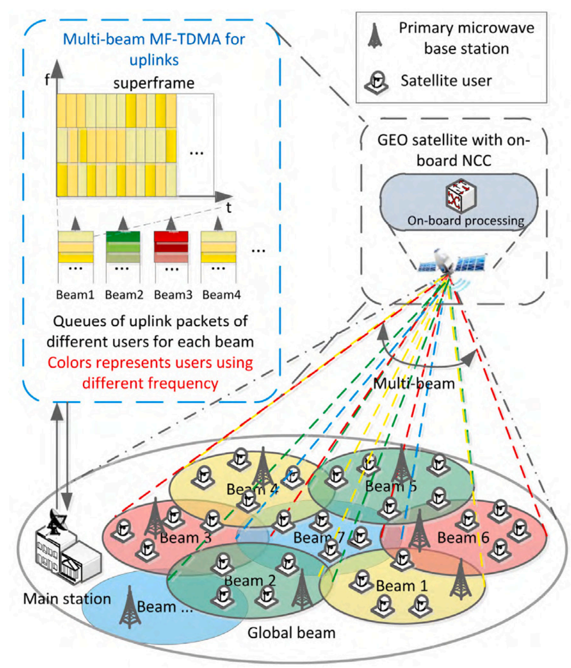

Fig. 7. Architecture of a multi-beam cognitive satellite network for uplinks. The system employs multi-beam technology and MF-TDMA to enhance frequency utilization. The GEO satellite uses one global beam for control channel and L beams for user capacity. Users are overlaid on a primary terrestrial network and reuse the same frequency band as secondary users.
Source: from Ref. [99].

navigation strategies at unsignalized intersections, developing Nash and cooperative game models that proved the existence of pure-strategy Nash equilibria and their optimality in cooperative scenarios.

An et al. [193] introduced a game theory-based control strategy for coordinating trajectory following and direct yaw control in four-wheel independently actuated autonomous vehicles. They created a two-agent non-cooperative game framework, using discrete-time Riccati equations and dynamic programming to derive control strategies. Xue et al. [194] proposed a robust differential game method for distributed collision avoidance in MAS, incorporating linear dynamics and disturbances. Their approach integrated trajectory optimization into a game framework, ensuring local and global robust Nash equilibria.

Bhatt et al. [195] identified cost and constraint function structures that transform dynamic games into constrained dynamic potential games, simplifying multi-agent trajectory optimization. Zhang

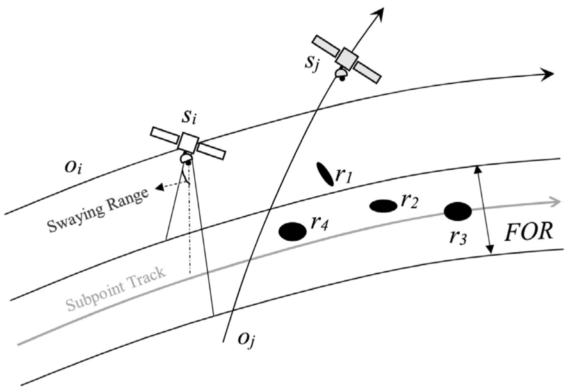

Fig. 8. Illustration of multi-satellite task allocation. The figure shows how satellite s_i can observe multiple targets within its field of regard, with $\{r_2, r_3, r_4\}$ representing the chronological task sequence $r_4 \rightarrow r_2 \rightarrow r_3$ for satellite s_i.
Source: from Ref. [190].

et al. [196] introduced a multi-level conflict game method to resolve priority conflicts in multi-agent pathfinding, addressing chain-like blocking issues and achieving conflict de-escalation. Giagkos et al. [197] provided insights into how game theory could be applied to solve UAV coordination problems in communication networks, optimizing flight paths to maintain connectivity while minimizing energy consumption. Pascarella et al. [198] exploited game theory for the assignment of inspection tasks and processing of optimal routes in drone coordination, achieving computational efficiency even with large numbers of agents in time-critical missions. Alexander Kuznetsov et al. [199] explored continuous optimization in multi-agent pathfinding, framing it as a game in Banach space and developing strategies to resolve conflicts and enhance collective movement, as illustrated in Fig. 9.

These studies advance the theoretical understanding of MASs interactions and enhance practical applications, improving efficiency and safety in real-world scenarios. The synergy between game theory and MASs continues to unlock new possibilities in path coordination and optimization.

5.5. Swarm-vs.-Swarm engagement

This subsection examines how game-theoretic models have been formulated to analyze strategic interactions between opposing agent groups, with focus on resource allocation, tactical decision-making, and coordination mechanisms in competitive multi-agent environments.

Hu et al. [200] studied behavior decision-making selection problems in UAV swarm missions under limited communication networks using evolutionary game analysis, revealing strategic adaptation patterns that enhance mission success rates in contested environments. Wu et al. [201] designed a community network for unmanned swarm cooperation and constructed a collaborative evolution model based on the multiplayer public goods game, demonstrating how altruistic behaviors can emerge and sustain in competitive aerospace scenarios.

Wang et al. [202] developed an algorithm based on the hedonic coalition formation game, enabling agents to form disjoint coalitions for different targets. This method mitigated the curse of dimensionality by reducing behavioral interactions, allowing a group of less capable agents to confront multiple stronger targets in hostile environments. Similarly, Wang et al. [203] addressed a 3D obstacle attack-defense scenario with N agents, as illustrated in Fig. 10, using a multi-population mean-field game approach combined with a GAN-based method. This approach effectively tackled the cooperative attack-defense evolution problem for large-scale agents in high-dimensional settings, reducing computational complexity and improving strategic decision-making.

Sun et al. [204] utilized a non-zero-sum game model with a multi-objective evaluation to tackle dynamic task allocation in MAS for autonomous underwater vehicles, incorporating a functional coordination mechanism to clarify decision-making and task allocation processes. Liu et al. [205] applied matrix game theory and an enhanced estimation of distribution algorithm to resolve target assignment in multi-UAV combat scenarios, framing the problem within a game-theoretic context and optimizing strategies for both offensive and defensive roles.

Ji et al. [206] modeled UAV swarm-vs.-swarm engagement using the Colonel Blotto game framework, addressing strategic resource allocation across battlefields. By applying Nash equilibrium, they identified optimal strategies ensuring no unilateral advantage, thus solving force allocation for UAV swarm engagements. Yao et al. [207] combined the Bayesian strong stackelberg algorithm with the win or learn fast algorithm through reinforcement learning to find equilibrium points in multi-agent Markov games, addressing moving target defense. Zhang et al. [208] treated swarm operations as a noncooperative game, introducing a graph-theoretic game model and employing a parallel maximum weight matching algorithm to achieve ε-Nash equilibrium, ensuring computational efficiency and suitability for swarm conflicts.

These studies illustrate the application of game theory in diverse strategic scenarios, from coalition formation to swarm operations and dynamic task allocation. Collectively, they advance the theoretical foundations of MASs in adversarial contexts and provide practical frameworks for developing intelligent autonomous systems capable of navigating modern conflict complexities.

5.6. Cooperative search and capture

This subsection explores how game theory has been applied to model strategic interactions in cooperative search and capture scenarios, with particular attention to area coverage optimization, target tracking strategies, and coordination mechanisms for multi-agent hunting operations.

Klima et al. [209] applied empirical game theoretic methods to study the strategic real-world problem of space debris removal, analyzing coordination challenges among multiple space agencies and commercial entities with different priorities and capabilities. Alsuwaidan et al. [210] explored the application of game theory in managing space debris, highlighting the importance of collaboration among stakeholders and proposing incentive mechanisms to overcome the tragedy of the commons in orbital environments. Ni et al. [211] applied an enhanced potential game theory with a modified binary log-linear learning algorithm to address cooperative search challenges in multi-UAV systems, improving efficiency and robustness while preventing aimless circling in zero-utility zones. Li and Duan [179] used a potential game framework and binary log-linear learning to solve cooperative search and surveillance problems for multiple UAVs, ensuring optimal coverage and integrating a consensus-based fusion algorithm for probability map construction.

Zheng et al. [212] developed a collaborative hunting algorithm using game theory and Q-learning for multi-agent hunting with a single escaper, deriving Nash equilibrium solutions for optimal strategies. Hua et al. [213] combined game theory, Q-learning, and the Apollonius circle to solve cooperative hunting for UAV clusters, dynamically adjusting strategies and payment matrices. Li et al. [214] established a multi-robot hunting model using extended cooperative game theory, considering task priorities, distances, and angles to the target.

Feng et al. [215] employed evolutionary task allocation and differential game theory for spacecraft cluster orbital games, using discrete particle swarm optimization and a hybrid evolutionary-newton method for autonomous operations. Ma et al. [216] applied differential game theory to design pursuit paths for the tethered space net robot (TSNR), as shown in Fig. 11, for active debris capture and removal. They established a game model with the tethered space net robot as the pursuer and the target as the evader, designed a cost function, and

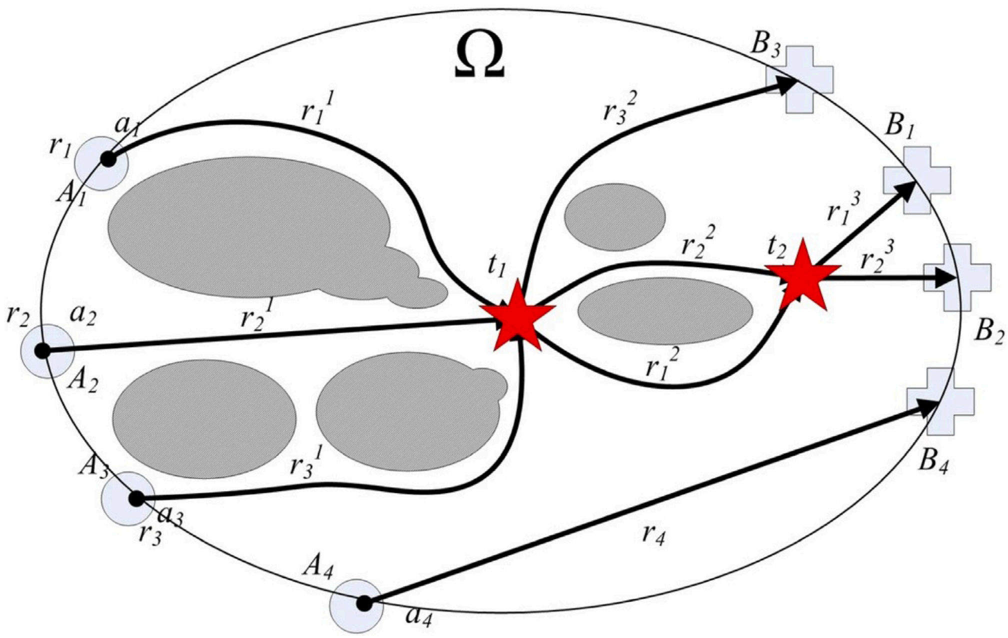

Fig. 9. Collisions description of agents a_1, a_2, and a_3 at time moments t_1, $t_2 \in [0, T]$ during the game. Agent a_4 is not a neighbor of other agents, while agent a_3 is a neighbor of a_1 and a_2 at time t_1 but not at t_2. Collision points are marked with stars, start locations with circles, and end locations with crosses. Agent routes are represented as state concatenations: $r_1 = r_1^1 r_2^2 r_3^3$, $r_2 = r_1^1 r_2^2 r_3^3$, $r_3 = r_1^1 r_2^2$. Obstacles are depicted as shaded subdomains.
Source: from Ref. [199].

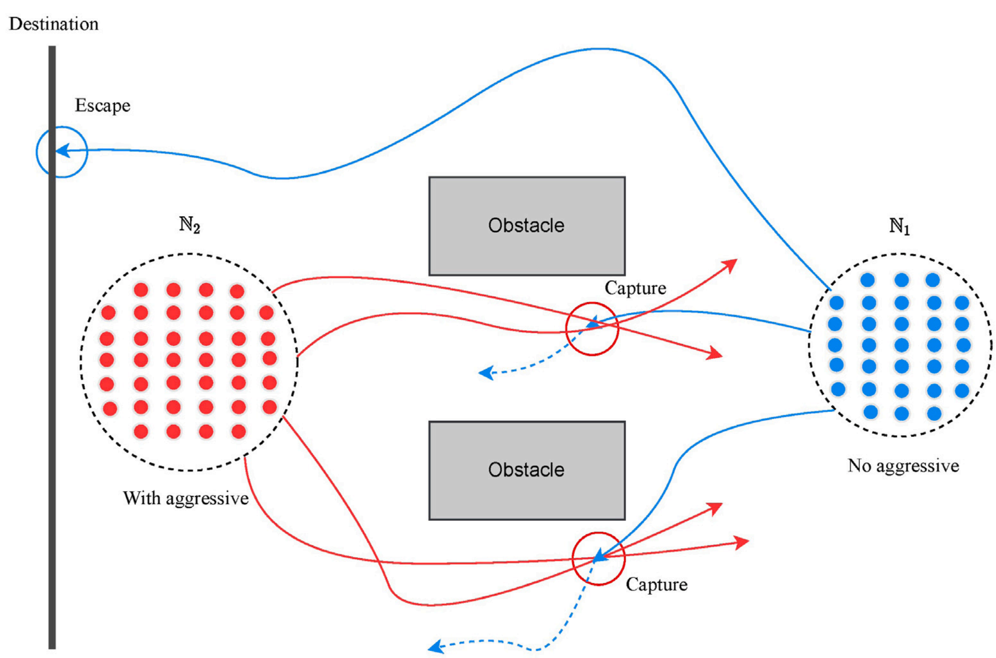

Fig. 10. Dynamic 3D multi-agent attack-defense scenario in an obstacle-rich environment. Blue (N_1) agents defend against red (N_2) attackers in a 3D space filled with obstacles. Red agents exhibit aggressive behavior, while blue agents focus on defense. Red arrows depict attackers' movements, and blue arrows indicate defenders' actions. A “Capture” occurs when an attacker is surrounded. Defenders aim to “Escape”, minimizing energy expenditure and reducing the risk of capture.
Source: from Ref. [203].

identified the optimal strategy pair, achieving a trajectory that ensures precise capture and stable formation.

The studies demonstrate the versatility of game-theoretic approaches in improving search efficiency, optimizing surveillance coverage, and refining hunting strategies in complex, dynamic environments. They underscore the importance of game theory in MASs and lay a strong foundation for future research in this domain.

5.7. Conclusion

In summary, this section has examined six distinct application scenarios where game theory provides valuable theoretical frameworks for modeling strategic interactions in MASs. From cooperative control and cooperative guidance to resource allocation, path coordination, swarm-vs.-swarm engagement, and cooperative search and capture,

Table 3
Characteristics of different scenarios and game-related approaches.

Scene	Characteristics	Game-theoretic formulations
Cooperative control	Reduce costs, ensure tracking consistency, conduct formation control, and make decisions during conflicts.	Cooperative game [159], Differential game [160], Potential game [161], Static game [170].
Cooperative guidance	Ensure cooperative pursuit and interception, swarm guidance and collision avoidance.	Markov game [172], Differential game [173], Evolutionary game [174], Cooperative game [175,176], Stackelberg game [177,178], Mean-field game [100].
Resource allocation	Optimize task allocation and resource coordination for cooperative benefits and efficiency.	Stochastic game [182,183], Evolutionary game [185], Non-cooperative game [184], Stackelberg game [181], Markov game [188].
Path optimization	Optimize data transmission paths, path-finding, collision avoidance, and priority conflict coordination.	Potential game [191], Non-cooperative game, Cooperative game [192], Differential game [194].
Swarm-vs.-swarm engagement	Collaborative attack-defense evolution of large-scale agents.	Hedonic coalition formation game [202], Colonel Blotto game [206], Mean-field game [203], Stackelberg game [207], Matrix game [205].
Cooperative search	Cooperative control, search, surveillance, area coverage, target tracking and encirclement.	Potential game [211], Cooperative game [214], Evolutionary game [215], Differential game [216].

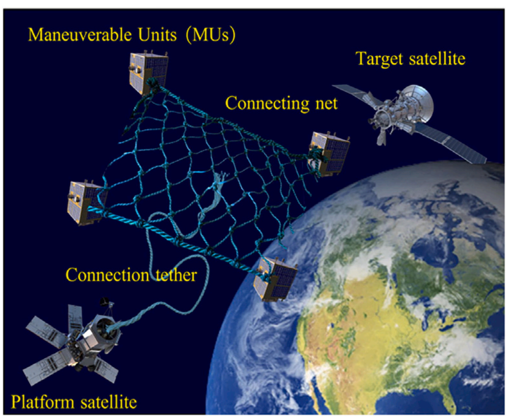

Fig. 11. Schematic representation of the TSNR system. The TSNR system comprises a platform satellite, multiple MUs, connecting nets, and tethers designed for active space debris capture and removal. The MUs are strategically controlled to form a net-like structure, enabling precise approach and engagement with the target satellite.
Source: from Ref. [216].

these frameworks offer rigorous mathematical foundations for analyzing agent behaviors and deriving optimal strategies under various constraints, as illustrated in Table 3.

The theoretical frameworks reviewed across these domains demonstrate the versatility and analytical power of game-theoretic approaches in addressing complex multi-agent coordination challenges. While these models show significant promise, their practical implementation faces constraints related to computational resources, communication bandwidth, and hardware limitations—particularly in complex scenarios such as swarm-vs.-swarm engagement. Despite the inherent complexity and cost challenges associated with physical deployment of swarm-vs.-swarm engagements, the aforementioned research primarily validates the effectiveness of game-theoretic strategies through advanced simulation platforms and rigorous mathematical modeling.

Future research will leverage recent advancements in distributed artificial intelligence, mobile edge computing, and high-bandwidth low-latency communication protocols such as 5G and satellite communications to gradually transition theoretical models to physical validation platforms [217]. Advancements in sensor technologies, including multi-target precise estimation capabilities and real-time environmental perception, will provide swarm agents with the necessary situational awareness to support real-time game-theoretic decision-making [218]. Additionally, optimizations in energy management and CPU performance, specifically tailored for UAV platforms, will ensure swarm agents can maintain their computational and operational capabilities over extended durations [219,220]. Through the integration of these technologies, future research will enable small-scale hardware-in-the-loop testing in controlled environments, progressively validating the practical feasibility and reliability of game theory in swarm-vs.-swarm engagements.

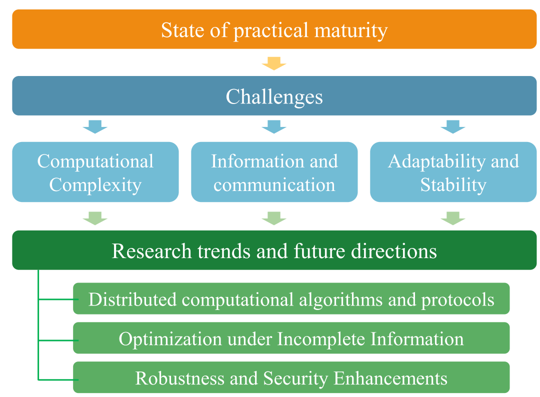

Fig. 12. Game-theoretic MASs: Maturity, challenges, and outlook.

A detailed examination of these challenges and promising research directions to address them will be presented in the following sections, where we explore computational complexity considerations, information constraints, and strategies for enhancing system robustness in multi-agent environments.

6. Discussion and future outlook

In the domain of MASs, game theory serves as a foundational tool for understanding and analyzing strategic interactions among autonomous agents. However, implementing game theory models in practice presents significant challenges that require thorough investigation. Concurrently, ongoing research aims to identify future trends and directions to advance the development and evolution of MASs. This section first evaluates the practical maturity of game-theoretic approaches across application domains, then analyzes underlying challenges that constrain real-world deployment, and finally highlights emerging research trends to bridge the gap between theory and practice, as illustrated in Fig. 12.

6.1. State of practical maturity

An intuitive evaluation of real-world readiness is provided through a maturity grading system based on three orthogonal dimensions:

Table 4
Practical maturity assessment of game-theoretic MAS approaches (V=validation level, S=scalability, E=environmental complexity).

Application	Maturity	Scale	Validation evidence
Resource allocation	V3-S2-E2	5–30	Spectrum allocation for UAV networks using Vickrey–Clarke–Groves mechanism design achieved field testing with 10 UAVs [99]. Satellite resource scheduling via cooperative bargaining games validated in hardware-in-the-loop simulations [188].
Cooperative control	V3-S1-E2	5–12	Multi-UAV formation flight tests using potential game reformulation of consensus algorithms demonstrated with 8–12 UAVs [168].
Path optimization	V3-S1-E2	2–10	Collision avoidance for UAVs in GPS-denied environments using potential games validated in outdoor flight tests with 5 UAVs [168].
Cooperative guidance	V2-S1-E2	2–8	Cooperative pursuit-evasion using differential games validated in hardware-in-the-loop simulations with quadrotors [173].
Adversarial interaction	V1–V2-S1-E3	2–5 (simulation)	Colonel Blotto games for resource allocation in swarm-vs.-swarm scenarios remain simulation-dominant with limited hardware-in-the-loop validation [206]. No public field tests of adversarial Nash equilibrium solvers exist due to safety constraints.
Cooperative search	V2-S1-E2	4–10	Area coverage using potential games validated in indoor robot experiments with 6–10 agents [211].

Validation level (V1–V4): V1 (pure simulation) → V2 (hardware-in-the-loop) → V3 (field experiments) → V4 (operational deployment). Grades reflect the highest validation tier with documented autonomous decision-making.
Scalability (S1–S3): S1 (2–15 agents, exact equilibrium solvable) → S2 (16–100 agents, approximate methods) → S3 (100+ agents, fully distributed rules). Grades reflect the largest agent count with physical validation.
Environmental complexity (E1–E3): E1 (static known) → E2 (dynamic predictable) → E3 (adversarial uncertain). Grades correspond to the most challenging validated environment type.

The maturity triplet (e.g., V3-S2-E2) concisely encodes an approach's validation status. Table 4 summarizes assessments for six application domains.

This maturity assessment reveals a distinct gradient across application domains. Resource allocation and cooperative control have achieved limited field validation at small scales (V3-S1 to V3-S2), whereas adversarial swarm interactions remain confined to simulation (V1–V2) due to computational and safety constraints. These observed maturity levels directly reflect the theoretical challenges examined in the following subsections, establishing a foundation for identifying pathways toward robust aerospace implementations.

6.2. Challenges

6.2.1. Computational complexity

In MASs, the computational complexity of game-theoretic models increases exponentially with the growth of state and strategy spaces as the number of agents rises [221,222], often resulting in NP-hard problems [223]. This curse of dimensionality complicates agents' learning and planning, as each agent's strategy is influenced by others, creating a complex network of strategic interactions [224]. Table 5 provides a comprehensive comparison of time, space, and communication complexities across different game-theoretic approaches, demonstrating how algorithmic efficiency scales with system parameters.

As shown in the table, the time and space complexity burdens vary significantly across methods, with some approaches (e.g., Shapley value with time complexity $O(N \cdot 2^N)$ and space complexity $O(2^N)$) becoming intractable even for moderate numbers of agents. The challenge is further intensified in strategy space optimization, particularly with continuous action spaces and complex payoff functions, hindering real-time solutions and limiting practical applications. Thus, MASs design must address computational constraints and seek more efficient algorithms, such as mean-field approximations that can achieve near N-independent complexity in certain structured scenarios, or evolutionary dynamics with linear space requirements.

6.2.2. Information and communication

The effectiveness of game-theoretic models in MASs relies on the assumption of perfect information, where all agents share common knowledge of strategies and utility functions [229]. This imposes strict demands on communication patterns and network topology. However, in practice, perfect information is often unattainable due to communication limits, perceptual errors, or intentional information concealment [230]. Table 5 further reveals that communication complexity varies dramatically across methods, highlighting the fundamental trade-off between information requirements and scalability [182].

Additionally, information security is critical [231]. Measures must protect sensitive data and prevent tampering or contamination during transmission [232]. High-communication methods (e.g., Bayesian Nash equilibrium) create larger attack surfaces due to extensive data exchange, while low-communication approaches (e.g., evolutionary games with $O(N)$ complexity) inherently reduce security vulnerabilities through localized interactions. Methods with minimal communication requirements often exhibit better robustness to information tampering and eavesdropping attacks. MASs design must balance communication efficiency with robust information security to ensure system stability and reliability.

6.2.3. Adaptability and stability

Adaptability and stability are vital for MASs performance under external perturbations and internal randomness [233–235]. Agents must adjust their strategies promptly and accurately in response to external changes or internal errors. Adversarial attacks, for instance, require online security assessments to mitigate potential risks [236].

Nonlinear interactions and strong coupling in MASs mean that one agent's strategy update can affect others, creating a non-stationary game environment [229]. Agents must adapt to others' strategic shifts to maintain effectiveness and system stability, increasing the complexity of learning and decision-making [237].

Table 5
Computational complexity comparison of game-theory-based methods.

Game type	Method name	Time complexity	Space complexity	Communication complexity	Core features
Non-cooperative	Pure-strategy Nash equilibrium solver [192]	$O(K^N)$	$O(K^N)$	$O(N^2)$	For static decision-making; high sensitivity to strategy space size.
	Mixed-strategy Nash equilibrium solver [205,225]	$O((NK)^{3.5})$	$O(NK)$	$O(N^2)$	Adapts to uncertain strategies; polynomial complexity in agent count.
	Correlated equilibrium solver [226]	$O(N^2K^2)$	$O(N^2K^2)$	$O(N \log N)$	Lower complexity than Nash equilibrium; suited for distributed allocation.
Cooperative	Shapley value allocation [165,227]	$O(N \cdot 2^N)$	$O(2^N)$	$O(N^2)$	Ensures fair payoff distribution; limited to small-scale scenarios.
	Nucleolus allocation [134]	$O(N^4 2^N)$	$O(2^N)$	$O(N^2)$	Robust to coalition changes; for constrained resource allocation.
	Hedonic coalition formation game [202,228]	$O(N^3 \log N)$	$O(N^2)$	$O(N \cdot C)$	Dynamic coalition adjustment; fits swarm-vs.-swarm engagement tasks.
Dynamic/ Stochastic	Differential game (HJB solver) [194]	$O(ND^2T)$	$O(D^2)$	$O(NT)$	Continuous-time interaction; used for real-time guidance.
	Markov game (Q-learning based) [182]	$O(NSA)$	$O(NSA)$	$O(NS)$	Adapts to dynamic environments; suitable for task scheduling.
Evolutionary/ Mean-field	Evolutionary game (Replicator dynamics) [185]	$O(N^2T)$	$O(N)$	$O(N)$	Low communication overhead; for large-scale swarm coordination.
	Mean-field Game [100]	$O(D^2T)$	$O(D^2)$	$O(1)$	Scales to ultra-large agent groups; near N-independent complexity.
	Public goods game with reputation [166]	$O(NT)$	$O(N)$	$O(N)$	Minimal complexity; supports collaborative task execution.
Incomplete information	Bayesian Nash equilibrium solver [170]	$O(N^3K\lvert\Theta\rvert)$	$O(NK\lvert\Theta\rvert)$	$O(N^2\lvert\Theta\rvert)$	Handles uncertainty; applicable to adversarial MASs.
Zero-sum	Colonel blotto game [206]	$O(N \cdot \mathrm{poly}(R, K))$	$O(NR)$	$O(N)$	Strategic resource distribution; for battlefield deployment.
Potential game	Log-linear learning [191]	$O(N \cdot T)$	$O(N)$	$O(N)$	Guarantees Nash convergence; used for path optimization.

Parameters: N = number of agents; K = size of discrete strategy set per agent; D = state dimension; T = number of iterations/convergence steps; S = size of state space; A = size of action space; $|\Theta|$ = size of type space; R = total resource amount; C = number of coalitions.
Complexity considerations: Actual computational requirements vary with algorithmic choices, problem instances, and implementation optimizations, reflecting practical rather than absolute theoretical limits.

Stability and adaptability are interdependent: stability ensures performance under disturbances, while adaptability enables flexible responses to environmental changes. Striking a balance is crucial—overemphasizing stability may lead to rigidity, while excessive adaptability may cause instability. MASs design must harmonize these aspects to ensure both stability and agility.

6.3. Research trends and future directions

In the realm of MASs, continuous exploration and advancement are unfolding across various dimensions. This subsection highlights key research trends and future directions poised to shape the development and evolution of MASs, focusing on distributed computational algorithms and protocols, optimization under incomplete information, and robustness and security enhancements.

6.3.1. Distributed computational algorithms and protocols

To address the dimensionality explosion in solving multi-agent problems with game-theoretic models, efficient distributed computational algorithms and protocols are essential for enabling collaborative work and task completion [228,238,239]. These algorithms and protocols often draw inspiration from natural collective behaviors, such as ant foraging patterns or bird flocking formations [240]. Natural systems, despite comprising numerous individuals with limited intelligence and localized information exchange, exhibit complex group behaviors through simple individual strategies [241]. This demonstrates MASs' ability to coordinate complex game-theoretic tasks without centralized control.

Distributed computing requires agents to design game strategies under partial observability, ensuring consistency with system objectives despite limited information exchange. This prevents task failure due to strategy evolution that could fragment MASs network topologies. Distributed computing avoids computational load dimensionality explosion, enabling large-scale collaboration and fostering collective intelligence as the number of agents increases [226,242]. Future research can further explore emergent complexity and its relationship with individual agents' simple game-theoretic strategies, aiming to discover more efficient and robust distributed computational methods and protocols.

6.3.2. Optimization under incomplete information

Reinforcement learning offers a powerful solution to challenges posed by incomplete information [243,244]. It enables agents to adapt and learn optimal strategies through interactions with the environment and other agents [245]. Agents can identify and understand peers' behavioral patterns, adjusting their game-theoretic strategies accordingly [246]. Developing strategy evaluation algorithms for agents under incomplete and asymmetric information is crucial for designing and training adaptive MASs game-theoretic models.

Balancing efficiency and security in agent communication strategies is vital. By limiting communication to necessary instances, the communication burden is reduced, and the risk of information tampering is minimized [247]. Additionally, investigating the impact of MASs network topology on system security aims to enhance robustness and game performance against misinformation, bias, and harmful information through superior network design [248,249].

Recent advancements in large language models (LLMs) offer new possibilities for MASs development [250]. Trained on extensive textual data, LLMs can respond to diverse queries, enhancing agent language communication mechanisms [251]. Integrating LLMs with game theory allows MASs to translate complex strategic concepts into executable plans and formulate targeted game strategies based on natural language descriptions of environments, objectives, and constraints.

6.3.3. Robustness and security enhancements

Future research will focus on optimizing MASs organizational structures to adapt to complex game environments and enhancing the robustness of game-theoretic algorithms to maintain stability against malicious attacks and information leaks [231]. Developing secure protocols and autonomous defense mechanisms will prevent malicious behaviors among agents [252–254]. Research will also integrate safety constraints into game theory, designing robust algorithms to handle uncertainties like sensor failures and communication disruptions, achieving fault tolerance and recovery strategies that adapt to environmental changes and maintain system stability [255]. To bolster data security, research will incorporate game algorithms with encryption and authentication mechanisms, such as homomorphic encryption, enabling strategy computation on encrypted data without decryption, safeguarding against breaches and unauthorized access while facilitating useful interactions [256].

In summary, distributed computational algorithms and protocols offer an efficient framework for collaborative problem-solving in MASs. Techniques for optimization under incomplete information enhance agents' decision-making in complex, real-world environments. Additionally, robustness and security measures ensure stable and secure system operations. These research directions are interconnected and mutually reinforcing, collectively defining a promising future for MASs advancement. Further exploration in these areas promises to develop more intelligent, efficient, and reliable MASs, driving technological progress and expanding applications in related fields, thereby providing robust solutions to real-world challenges.

The discussed research trends and future directions outline a roadmap for transforming MAS theory into practical applications. These efforts are essential not only for overcoming current obstacles but also for advancing MASs toward more intelligent, efficient, and reliable applications in diverse, dynamic scenarios, ultimately shaping the future of this field.

7. Conclusion

This paper offers a comprehensive analysis of the fundamental elements and classifications of game theory and MASs, examining their interconnections and evaluating the application and effectiveness of game theory within MASs. The findings demonstrate that game theory serves as a robust theoretical framework, enabling the development of versatile and practical MASs with high efficacy. In diverse scenarios such as cooperative control, cooperative guidance, resource allocation, path coordination, and swarm-vs.-swarm engagement, game theory has demonstrated significant value in addressing practical MAS challenges.

Nevertheless, the paper identifies critical challenges in the field, including computational complexity, information and communication constraints, and issues related to adaptability and stability. Future research should prioritize the development of distributed computational algorithms and protocols, optimize strategies under incomplete information, and enhance system robustness and security. With continuous technological advancements, the field is well-positioned to deliver more efficient and intelligent solutions to tackle the complex challenges of real-world applications.

CRediT authorship contribution statement

Ruizhe Feng: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shuangxi Liu:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Wei Huang:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Tuo Han:** Supervision, Resources, Project administration, Conceptualization. **Binbin Yan:** Supervision, Resources, Project administration, Methodology, Conceptualization. **Zhongwei Wang:** Supervision, Resources, Project administration, Methodology, Conceptualization. **Yaobin Niu:** Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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